

# Bike-Sharing Demand Analysis

Executive Findings and Operational Recommendations

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**Data granularity:** Daily and hourly observations

**Analytical tools:** Python and JMP

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## Report Purpose

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This report summarizes the main findings of the bike-sharing demand analysis for a management audience.

## 1 Executive Summary

This report evaluates historical bike-sharing rental activity to support system-level operating and planning decisions. The analysis uses 731 daily observations and 17,379 hourly observations covering January 2011 through December 2012. Daily and hourly source files were audited, cleaned on separate working copies, reconciled, and analyzed independently before the findings were combined for management reporting.

The data were generally suitable for analysis. No missing cells or duplicate records were identified, rental-count identities were internally consistent, and the available hourly rentals reconciled exactly with the corresponding daily totals. Two data-quality concerns were formally documented: 165 expected date-hour combinations were absent across 76 dates, and humidity was recorded as zero on March 10, 2011. The original records were retained rather than imputed. Sensitivity checks showed that the principal casual, registered, and total peak hours remained unchanged when the hourly analysis was restricted to dates with complete 24-hour coverage.

Average daily rentals were 64.4% higher in 2012 than in 2011, and registered users accounted for 84.5% of the observed increase. Average demand in 2012 exceeded the corresponding 2011 level in all 12 calendar months. These results show that rental activity was substantially higher in 2012. However, two years of data are not enough to confirm a long-term growth trend or explain why demand increased.

Demand timing and user mix differed across segments and day types. Registered users represented 86.8% of working-day rentals and recorded pronounced demand increases around 08:00 and 17:00, with their highest average hourly volume occurring at 17:00. Casual demand followed a broader daytime pattern and peaked at 14:00. Average weekend casual demand was 124.3% higher than the Monday-to-Friday average. On non-working days, total demand peaked earlier, at 13:00. These patterns show why separate operating schedules are needed for different user segments and day types.

Weather was also associated with substantial variation in completed rentals. Historical average daily demand was approximately 4,877 rentals under clear or partly cloudy conditions, 4,036 under misty or cloudy conditions, and 1,803 under light rain or snow. The light-rain-or-snow average was therefore approximately 63.0% lower than the clear-or-partly-cloudy average. The preferred statistical models continued to show substantial adjusted weather differences after accounting for year, season, calendar structure, temperature, humidity, windspeed, and temporal dependence. However, light rain or snow was observed on only 21 days, and no daily observations were available for the most severe weather category.

Separate time-series regression models were selected for total, casual-user, and registered-user demand. Each preferred model incorporated daily and weekly autoregressive error components. The models substantially reduced the strong short-term residual autocorrelation observed in the ordinary least squares baselines, although broader Ljung-Box tests indicated that some serial dependence remained, particularly for casual-user demand. The models help explain historical

demand patterns after adjustment for the included factors. They should not be interpreted as causal estimates or validated forecasts.

The analysis supports five management priorities. First, operating readiness should receive greater attention before the main working-day registered-user demand periods. Second, separate time-of-day operating templates should be used for working and non-working days and for registered and casual demand patterns. Third, weekend and holiday planning should recognize the substantially higher casual-user activity around midday and early afternoon. Fourth, short-term weather information should adjust the relevant operating template rather than trigger fixed or automatic capacity reductions. Fifth, demand volume and user composition should be monitored routinely before long-term fleet or capacity commitments are made.

The analysis identifies the periods when system-wide demand was highest. It does not show whether capacity was sufficient, which stations needed intervention, how many bicycles should be moved, or whether fleet expansion would be financially justified. These decisions require station-level bicycle and dock availability, inventory, origin–destination, stockout, service-interruption, operating-cost, and unmet-demand data.

Overall, the project provides a validated and reproducible historical baseline for system-level planning. Its main contribution is a practical framework for aligning operating plans with observed demand patterns while remaining within the limits of the available data.

## 2 Business Context and Decision Questions

### 2.1 Business Context

Bike-sharing systems must balance service availability with demand that changes across seasons, day types, hours, user segments, and weather conditions. Poorly timed capacity allocation can reduce bicycle availability during high-demand periods while leaving excess capacity underused at other times. This report analyzes validated daily and hourly rental records covering January 2011 through December 2012. Demand is evaluated for total rentals and separately for casual and registered users because the two segments follow different usage patterns and may therefore require different operational responses. The analysis is intended to support system-level decisions related to service reliability, fleet allocation, rebalancing schedules, weekend capacity, and weather-sensitive planning. It does not attempt to estimate causal effects or optimize individual stations because the available data describe aggregated demand rather than station-level operations.

### 2.2 Decision Questions

The analysis addresses five decision questions:

1. How did daily demand change between 2011 and 2012, and which user segment contributed most to that growth?
2. How does demand vary across months, working and non-working days, weekdays, and hours of the day?
3. How do the demand patterns of casual and registered users differ?
4. How is observed demand associated with weather conditions after accounting for calendar and environmental factors?

5. What operational actions are supported by the combined descriptive and statistical evidence?

## 3 Data Scope and Reliability

### 3.1 Data Coverage

The analysis uses two complementary datasets covering the period from January 1, 2011, through December 31, 2012. The daily dataset contains 731 observations, with one record for each calendar day. The hourly dataset contains 17,379 observations and provides a more detailed view of within-day demand patterns. Both datasets report total rentals and separate demand into casual and registered users. They also include calendar and environmental variables such as season, month, weekday, holiday status, working-day status, weather category, temperature, humidity, and windspeed. Daily observations were used for annual, monthly, weekday, weather, and time-series analyses. Hourly observations were used to examine differences in demand patterns across hours of the day. The daily data were used to examine broader demand patterns, while the hourly data were used to identify within-day peaks.

### 3.2 Data-Quality Assessment

A structured data audit was completed before cleaning, analysis, and modeling. No missing values or exact duplicate records were identified in either dataset. Rental counts were non-negative, categorical variables remained within their documented ranges, and the identity between total, casual, and registered rentals was satisfied in every observation.

Daily totals were also reconciled with the corresponding aggregation of hourly records, providing an important cross-dataset validation check. The daily date sequence was complete and correctly ordered across the two-year analysis period. The hourly dataset covered all 731 dates, although some dates had incomplete hourly coverage.

Two data-quality issues were retained in the project issue register. First, 165 expected date-hour combinations were absent from the hourly dataset, meaning that hourly coverage was not fully complete.

Second, a humidity value of zero was recorded on March 10, 2011. These observations were not automatically replaced because the available data did not provide sufficient evidence for a defensible corrected value.

The identified issues were documented and considered during interpretation. Neither issue invalidated the main analysis, but both limit claims of perfectly complete hourly coverage and fully error-free environmental measurements.

### 3.3 Analytical Scope

The project combines descriptive analysis, cross-tool regression validation, time-series modeling, and explanatory visualization. Descriptive analyses were used to identify demand growth, seasonality, user-segment differences, hourly peaks, day-type patterns, and weather-related variation. Ordinary least squares models were estimated in JMP, and the total-demand OLS baseline was reproduced in Python as a cross-tool validation step. Because the baseline residuals showed

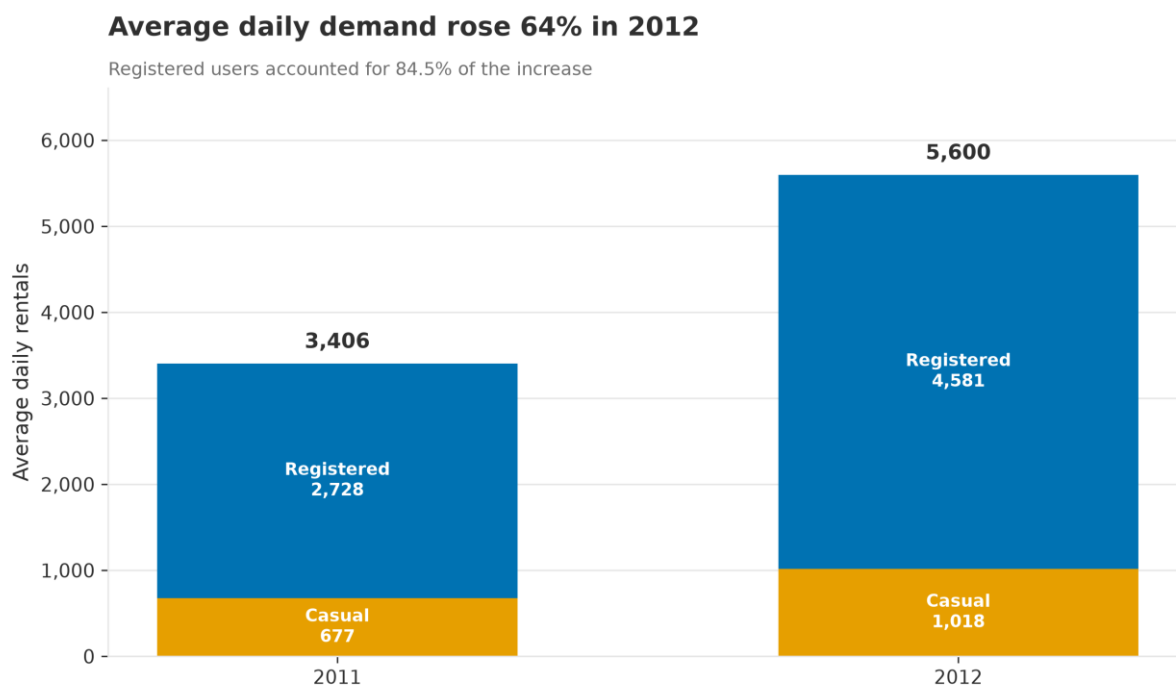


positive temporal autocorrelation, preferred daily models were subsequently estimated in Python with daily and weekly autoregressive structure. The results should be interpreted as adjusted associations rather than causal effects. The data are observational and do not contain randomized interventions or a formal identification strategy. In addition, the available records describe system-level demand and do not include station-level inventory, trip origins and destinations, pricing decisions, maintenance events, or operating costs. The report can support system-level planning and scheduling, but it cannot optimize individual stations, estimate causal policy effects, or provide a complete financial business case.

## 4 Key Findings

### 4.1 Demand Growth Was Concentrated Among Registered Users

Average daily demand increased from 3,406 rentals in 2011 to 5,600 rentals in 2012, representing growth of 64.4%. The increase was observed in both user segments, but it was concentrated primarily among registered users. Average daily registered-user demand rose from 2,728 to 4,581 rentals, an increase of approximately 1,853 rentals per day. Average casual-user demand increased from 677 to 1,018 rentals, adding approximately 341 rentals per day. Registered users therefore accounted for 84.5% of the total increase in average daily demand.



Average daily rentals, 2011–2012. Descriptive comparison.

Figure 1. Average daily demand growth and user-segment contribution, 2011–2012.

Source: Author's analysis of daily bike-sharing records. Values are descriptive averages.

Registered users also represented the majority of demand in both years, accounting for 80.1% of rentals in 2011 and 81.8% in 2012. Registered users already represented most rentals, and they also contributed most of the increase in 2012. Registered-user demand should therefore remain a central consideration in system-wide operating plans. These comparisons are descriptive. They establish the scale and composition of the observed increase but do not identify the factors that caused demand to grow between the two years.

## 4.2 Demand Was Higher in 2012 and Varied Across the Calendar Year

Monthly averages show that demand growth in 2012 was broad-based rather than limited to a small number of peak months. Average daily rentals in 2012 exceeded the corresponding 2011 level in every calendar month. Demand increased from the winter months into the warmer part of the year in both years, although the timing of the annual peak differed. In 2011, the highest monthly average occurred in June, at 4,784 rentals per day. In 2012, demand continued rising later into the year and reached its highest monthly average in September, at 7,286 rentals per day. Demand then declined toward the end of the year in both periods.

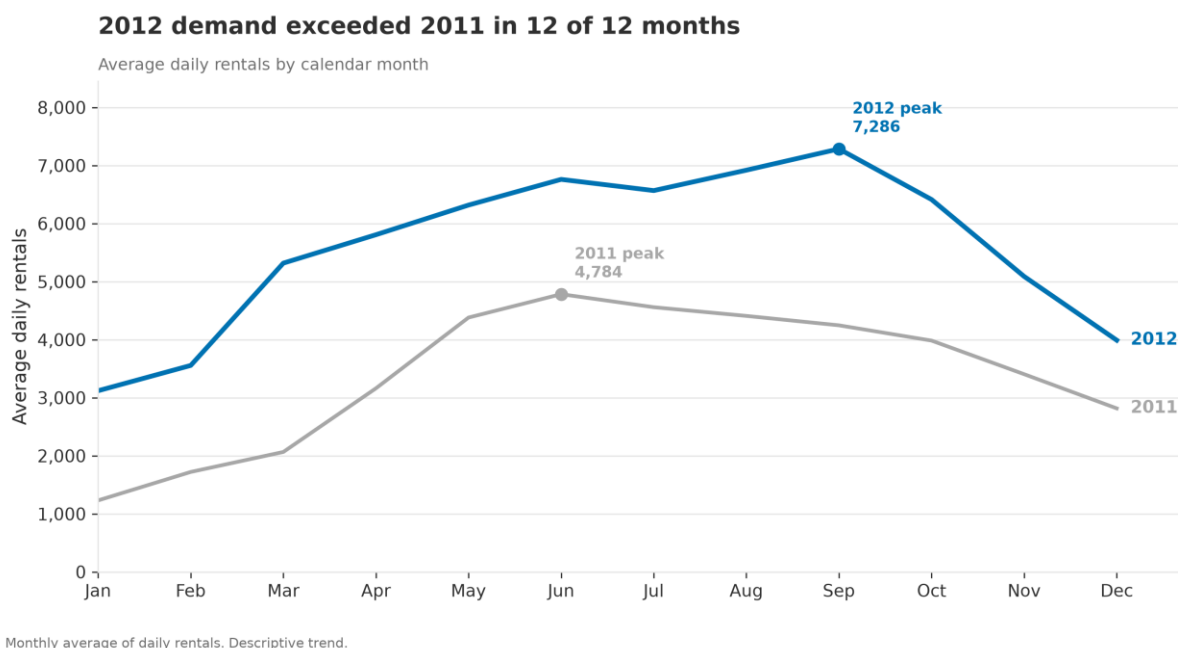


Figure 2. Monthly average daily demand in 2011 and 2012.

Source: Author's analysis of daily bike-sharing records. Monthly values are descriptive averages.

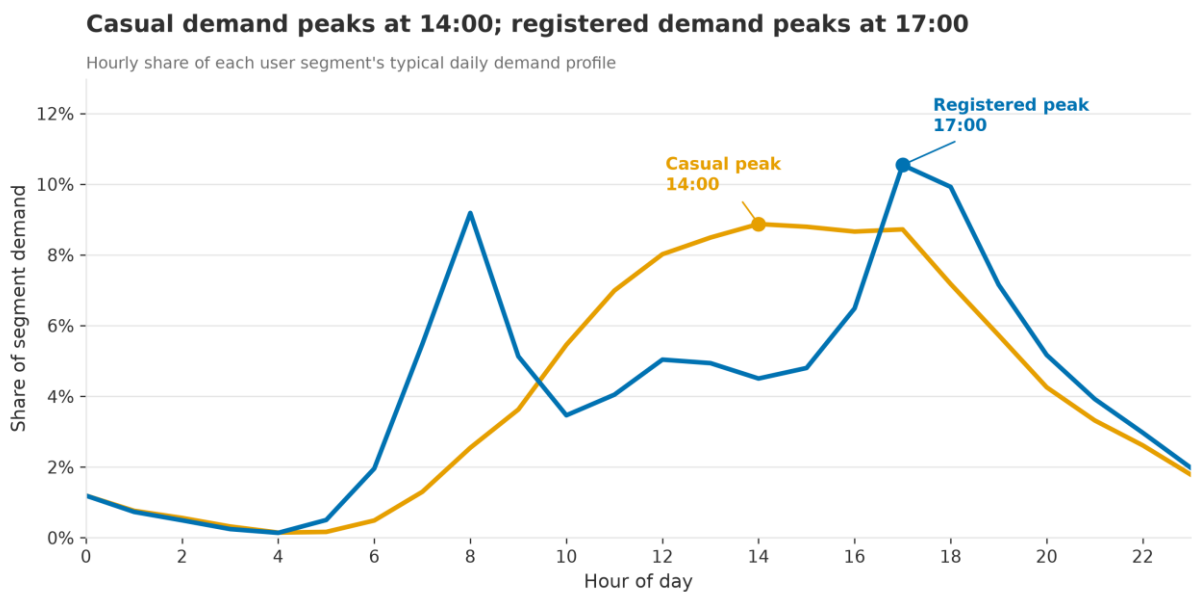
Because demand differed both between years and across months, a single year-round operating assumption would not reflect the observed demand pattern. Operating plans should account for both the higher 2012 demand level and the variation observed across the year.

These findings describe observed monthly averages. They do not constitute a demand forecast and do not identify the causes of the difference between the two years.

### 4.3 Casual and Registered Users Followed Different Hourly Patterns

Hourly demand profiles differed substantially between casual and registered users. Casual demand increased gradually through the morning, remained elevated during the middle of the day, and reached its highest average level at 14:00. Registered demand followed a more concentrated pattern, with a pronounced morning increase around 08:00 and its highest average level at 17:00.

The overall average peak for casual users was approximately 75.6 rentals per hour at 14:00. Registered demand reached approximately 387.2 rentals per hour at 17:00, while total system demand also peaked at 17:00 at approximately 461.5 rentals per hour.



Based on average observed demand by hour. Peak hours remained stable in the hourly-data sensitivity check.

Figure 3. Hourly demand profiles for casual and registered users.

Source: Author's analysis of hourly bike-sharing records. Profiles show each segment's hourly share of typical daily demand.

The registered-user pattern resembles morning and evening commuting activity, while casual demand is spread more broadly across the daytime. Trip purpose was not directly observed, so these interpretations should be treated as operationally useful patterns rather than confirmed behavioral explanations.

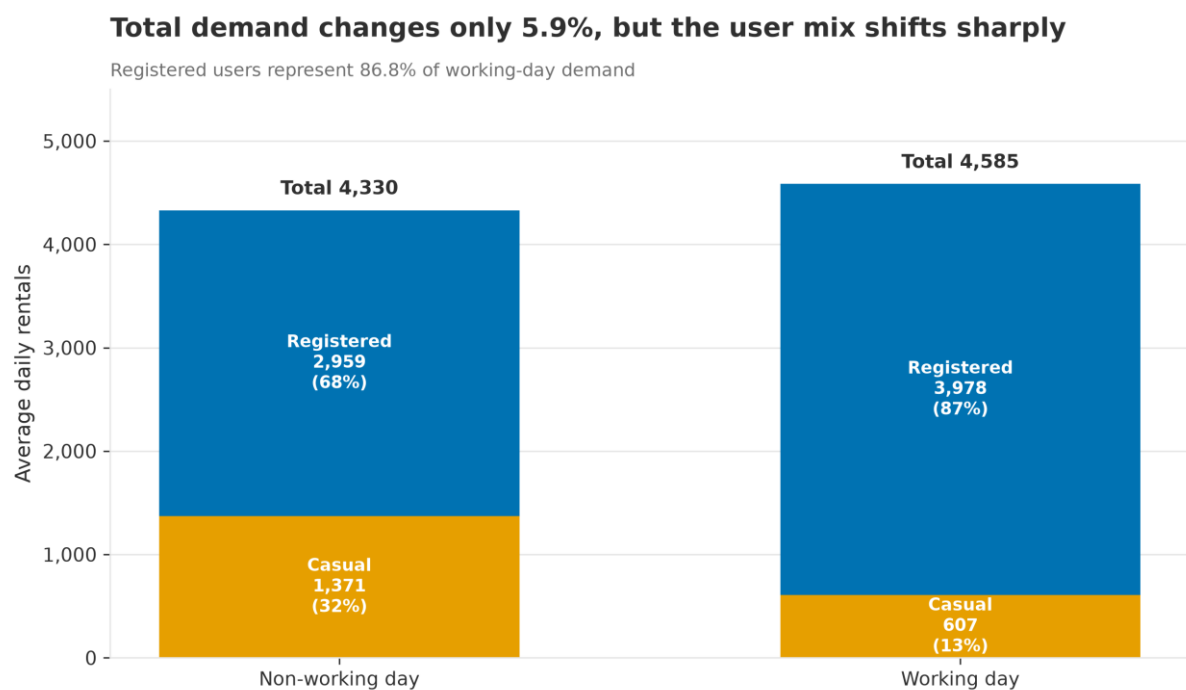
The peak-hour results were unchanged when the analysis was repeated using only dates with complete hourly coverage. Casual demand continued to peak at 14:00, while registered and total demand continued to peak at 17:00. This sensitivity check indicates that the identified peak hours were not driven by the missing date-hour combinations in the hourly dataset.

Hourly planning should distinguish between registered and casual demand rather than relying only on the total-demand profile. Operating preparation should focus on the registered-user morning and evening peaks and on the broader casual-user demand period around midday and the afternoon.

#### 4.4 Total Demand Concealed a Major Working-Day Shift in User Composition

Average total demand differed only moderately between working and non-working days. Working days averaged 4,585 rentals, compared with 4,330 rentals on non-working days—a difference of approximately 5.9%. Despite this relatively small difference in total volume, the composition of demand changed substantially.

Registered users averaged 3,978 daily rentals on working days and represented 86.8% of working-day demand. On non-working days, registered demand averaged 2,959 rentals and accounted for 68.3% of the total. Casual demand moved in the opposite direction, averaging 1,371 rentals on non-working days but only 607 rentals on working days. The casual share therefore declined from 31.7% on non-working days to 13.2% on working days.



Average daily rentals by working-day status. Descriptive comparison.

Figure 4. Average daily demand and user composition by working-day status.

Source: Author's analysis of daily bike-sharing records. Values are descriptive averages and shares.

Similar total rental volumes can hide important differences in user composition. Planning based only on total demand would miss the stronger registered-user share on working days and the larger casual-user share on non-working days.

Working and non-working days should therefore use different operating profiles. Working-day planning should emphasize reliability for registered users, while non-working-day planning should account for the larger casual-user component.

The comparison is descriptive. In this dataset, non-working days combine weekends and official holidays, so the category should not be interpreted as a pure weekend effect.

## 4.5 Casual Demand Was Concentrated on Weekends

Casual-user demand varied sharply across the week. Average daily casual rentals reached 1,465 on Saturdays and 1,338 on Sundays, substantially exceeding the levels observed from Monday through Friday.

Across the two weekend days, casual demand averaged approximately 1,402 rentals per day. The corresponding weekday average was approximately 625 rentals per day. Weekend casual demand was therefore 124.3% higher than weekday casual demand.

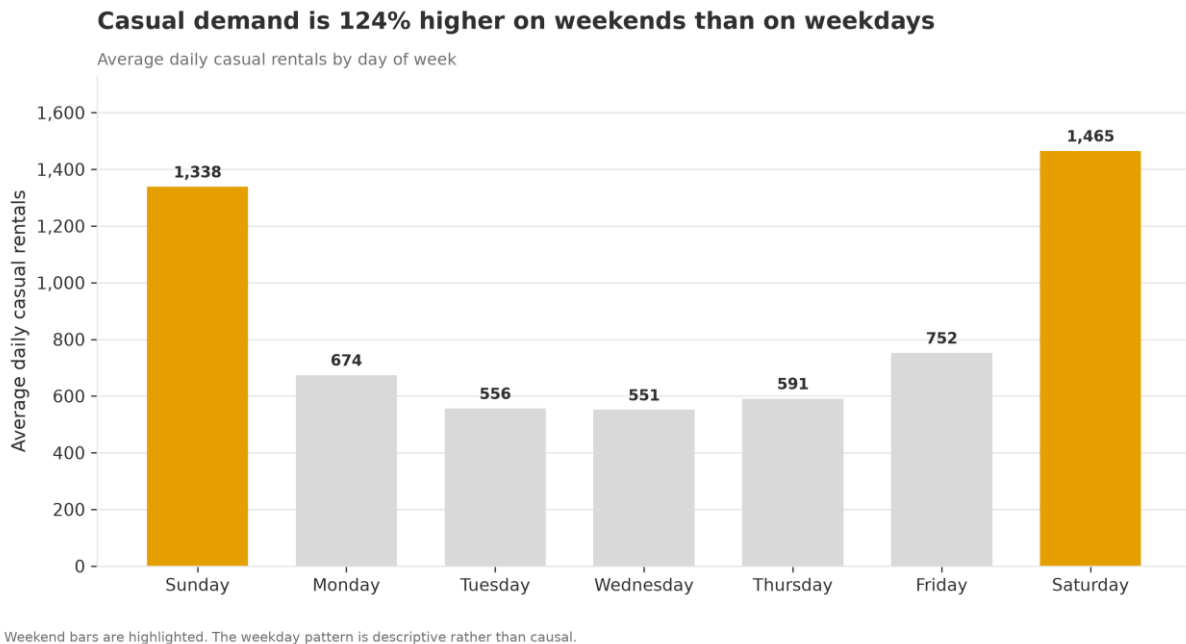


Figure 5. Average daily casual-user demand by day of week.

Source: Author's analysis of daily bike-sharing records. Weekend refers to Saturday and Sunday; values are descriptive averages.

Among weekdays, Friday recorded the highest average casual demand at 752 rentals per day, while Wednesday recorded the lowest at 551. The pattern suggests that casual demand begins to strengthen before the weekend but remains most concentrated on Saturdays and Sundays.

Weekend operating plans should give greater attention to casual demand during the daytime and afternoon. The available data do not show whether additional fleet capacity is required.

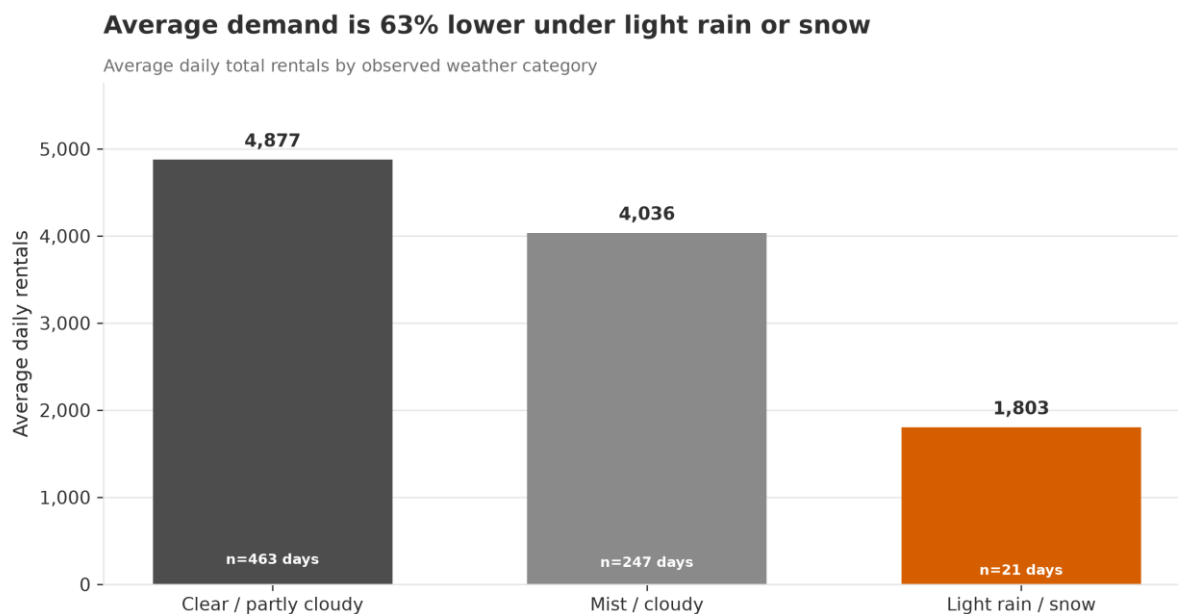
Weekend schedules should differ from working-day schedules because both the user mix and the timing of demand change.

These comparisons are descriptive and should not be interpreted as a causal weekend effect. Day-of-week averages may also reflect differences in season, weather, holidays, and other conditions that vary across the observed dates.

## 4.6 Adverse Weather Was Associated with Substantially Lower Demand

Observed demand declined as weather conditions became less favorable. Average daily demand was 4,877 rentals under clear or partly cloudy conditions and 4,036 rentals under misty or cloudy conditions.

Under light rain or snow, average daily demand fell to 1,803 rentals. This was approximately 63.0% lower than the average observed under clear or partly cloudy conditions. Lower demand was visible in both the casual and registered user segments.



Descriptive association; not a causal estimate. No daily observations were available for the heavy rain / snow / fog category.

Figure 6. Average daily total demand by observed weather condition.

Source: Author's analysis of daily bike-sharing records. Values are descriptive averages; sample sizes are shown in the figure.

This difference is large, but the adverse-weather sample was small. Clear or partly cloudy conditions were observed on 463 days, misty or cloudy conditions on 247 days, and light rain or snow on only 21 days. No daily observations were available for the most severe weather category.

Weather information should therefore be considered when adjusting short-term operating plans. On days with light rain or snow, expected system activity may be materially lower, while favorable-weather days require preparation for substantially higher demand.

These results describe associations in the observed data and do not establish a causal weather effect. Weather categories may also coincide with seasonal, calendar, and other environmental differences.

## 5 Statistical Evidence

### 5.1 Modeling Approach

Daily demand was modeled separately for total rentals, casual users, and registered users. Estimating separate models was necessary because the descriptive analysis showed that the two user segments followed different calendar and hourly demand patterns.

Ordinary least squares models were first estimated in JMP using year, season, calendar status, weather category, temperature, humidity, and windspeed as explanatory variables. The total-

demand OLS model was then reproduced in Python, where its coefficient estimates, model fit, and residual diagnostics agreed with the JMP results. This cross-tool comparison provided a validation check before the time-series analysis was extended.

The OLS residuals showed positive temporal autocorrelation, indicating that observations on adjacent days could not be treated as fully independent. Candidate time-series regressions were therefore estimated in Python using three progressively structured approaches: an OLS baseline, a model with daily AR(1) errors, and a model combining daily AR(1) errors with a weekly seasonal AR(1) component using a seven-day period.

Calendar sensitivity models were also evaluated. The main total-demand and registered-user specifications used working-day status, while an alternative specification replaced this variable with individual weekday and holiday effects. This more detailed calendar structure was retained for casual demand because it provided a better balance of model fit and interpretation for that segment.

Preferred models were selected separately for each response using convergence, AIC and BIC, residual autocorrelation, Durbin–Watson and Ljung–Box diagnostics, model simplicity, and operational interpretability. No observations were deleted or imputed for the main models. The final models were designed to estimate adjusted relationships while accounting for temporal dependence; they were not intended to establish causality or serve as maximally complex forecasting models.

## 5.2 Selected Models

The preferred model specifications differed in their treatment of calendar effects but shared the same temporal error structure. All three models used the complete 731-day series and included year, season, weather category, temperature, humidity, and windspeed as common explanatory variables.

The total-demand and registered-user models represented calendar status using working-day status. The casual-user model instead retained separate weekday and holiday effects, reflecting the more detailed weekly variation identified for this segment.

All three preferred models combined a daily AR(1) error term with a weekly seasonal AR(1) component using a seven-day period. Each model converged successfully. This structure was selected because it improved model fit and residual behavior while remaining relatively simple to interpret.

*Table 1 Preferred daily time-series regression models*

| Response         | Calendar specification      | Temporal error structure                      | AIC       | BIC       |
|------------------|-----------------------------|-----------------------------------------------|-----------|-----------|
| Total rentals    | Working-day status          | Daily AR(1) + weekly seasonal AR(1), period 7 | 11,718.59 | 11,782.92 |
| Casual users     | Weekday and holiday effects | Daily AR(1) + weekly seasonal AR(1), period 7 | 10,541.41 | 10,633.30 |
| Registered users | Working-day status          | Daily AR(1) + weekly seasonal AR(1), period 7 | 11,318.95 | 11,383.27 |

*Note: All models also include year, season, weather category, temperature, humidity, and windspeed. AIC and BIC are comparable only among candidate models estimated for the same response.*

Source: Author's preferred-model summary generated in Python following cross-tool validation of the OLS baseline in JMP.

### 5.3 Main Adjusted Relationships

After adjustment for the included calendar, weather, environmental, and temporal factors, the model results remained broadly consistent with the descriptive findings.

After adjustment, estimated daily demand in 2012 was approximately 2,008 rentals higher than in 2011. The corresponding adjusted differences were approximately 277 additional casual rentals and 1,712 additional registered rentals per day. The registered-user segment therefore remained the primary source of the year-to-year demand difference after the other modeled factors were held constant.

Calendar effects also differed by user segment. In the total-demand model, working days were associated with approximately 162 more rentals than non-working days. The corresponding adjusted difference for registered users was substantially larger, at approximately 835 additional rentals per working day. Casual demand was modeled using separate weekday and holiday effects rather than a single working-day indicator. In that model, Saturday and Sunday had the strongest positive adjusted deviations, while Monday through Friday were below the adjusted grand mean. Adjusted casual demand was also approximately 396 rentals higher on holidays than on non-holidays.

Weather remained strongly associated with demand after adjustment. The estimated difference between clear or partly cloudy conditions and light rain or snow was approximately 1,901 total rentals per day. The corresponding adjusted differences were approximately 333 casual rentals and 1,591 registered rentals.

Temperature was positively associated with all three demand measures. A one-degree Celsius increase was associated with approximately 112 additional total rentals, 40 additional casual rentals, and 74 additional registered rentals per day, holding the other modeled variables constant. Higher humidity and windspeed were associated with lower adjusted demand across all three models.

Table 2. Selected adjusted relationships from the preferred daily models

| Adjusted relationship                             | Total rentals | Casual users           | Registered users |
|---------------------------------------------------|---------------|------------------------|------------------|
| 2012 compared with 2011                           | +2,008        | +277                   | +1,712           |
| Working day compared with non-working day         | +162          | Not directly estimated | +835             |
| Clear/partly cloudy compared with light rain/snow | +1,901        | +333                   | +1,591           |
| Temperature: increase of 1°C                      | +111.9        | +39.5                  | +74.0            |
| Humidity: increase of 1 percentage point          | -16.6         | -5.6                   | -11.4            |
| Windspeed: increase of 1 original-scale unit      | -31.5         | -14.8                  | -17.7            |

Note: Values are adjusted differences in estimated daily rentals. Positive values indicate higher adjusted demand and negative values indicate lower adjusted demand. Category contrasts were calculated from sum-coded coefficients. The casual-user model used weekday and holiday effects instead of working-day status. Pairwise confidence intervals are not shown.

Source: Author's preferred time-series regression estimates generated in Python. The OLS baseline was cross-validated against JMP results.



Seasonal relationships were less uniform across the user segments. Spring was below the adjusted grand mean in all three models, while summer was positively associated with total and casual demand. Several fall and registered-user seasonal coefficients were not statistically distinguishable from zero. The year, calendar, and weather results were more consistent across the models than the seasonal results.

The estimates in this section are model-adjusted associations rather than causal effects. Categorical contrasts were derived from sum-coded coefficients, and their interpretation differs from a conventional reference-category model.

## 5.4 Residual Diagnostics

The baseline OLS models showed positive temporal autocorrelation, indicating that daily observations contained systematic dependence not captured by the explanatory variables alone. This result supported the use of time-series regression rather than relying exclusively on the conventional OLS independence assumption.

The preferred daily and weekly autoregressive structures substantially reduced the most immediate residual dependence. In the final models, residual lag-1 correlations ranged from  $-0.043$  to  $0.012$ , while residual lag-7 correlations ranged from  $-0.042$  to  $-0.010$ . Durbin–Watson statistics were also close to 2.0 for all three responses, indicating that the strong positive first-order autocorrelation observed in the baseline models had largely been removed.

*Table 3. Residual diagnostics for the preferred daily models*

| Response         | Residual lag-1 correlation | Residual lag-7 correlation | Durbin–Watson | Ljung–Box p-values: 7 / 14 / 30 lags |
|------------------|----------------------------|----------------------------|---------------|--------------------------------------|
| Total rentals    | $-0.0428$                  | $-0.0101$                  | 2.0847        | 0.0117 / 0.0011 / 0.0057             |
| Casual users     | 0.0124                     | $-0.0416$                  | 1.9702        | <0.0001 / <0.0001 / <0.0001          |
| Registered users | $-0.0259$                  | $-0.0122$                  | 2.0513        | 0.1532 / 0.0169 / 0.0543             |

*Note:* Residual correlations are based on standardized one-step-ahead forecast errors. Ljung–Box entries report p-values at 7, 14, and 30 lags. Values below 0.05 indicate evidence of remaining residual serial dependence.

*Source:* Author’s residual diagnostics for the preferred time-series regression models generated in Python.

The registered-user model had the largest daily error-persistence parameter, at 0.517, while the casual-user model had the largest weekly error-persistence parameter, at 0.252. These parameters describe dependence remaining in the model errors after adjustment for the fixed effects; they do not represent direct persistence in observed rental demand.

Broader residual diagnostics were less uniform. Ljung–Box tests remained statistically significant at the 7-, 14-, and 30-lag horizons for total demand. The casual-user model showed the clearest remaining residual dependence, with p-values below 0.0001 at all three horizons. Diagnostics for registered demand were more favorable, although the 14-lag test remained statistically significant.

The autoregressive models reduced residual dependence, but they did not remove it completely. This limitation does not invalidate the adjusted associations reported in the analysis, but it means that the models should not be presented as fully optimized forecasting systems. A forecasting model

would require out-of-sample validation and possibly additional temporal, nonlinear, or event-related terms.

## 6 Operational Implications

### 6.1 Registered-User Operations

Registered users account for most of the system's rental activity. They represented more than four-fifths of total demand and accounted for 84.5% of the observed increase in average daily rentals between 2011 and 2012. Operating plans should therefore give particular attention to registered-user demand.

Registered-user demand was especially concentrated on working days. They represented 86.8% of working-day demand, and the preferred registered-user model estimated approximately 835 additional rentals on working days compared with non-working days after adjustment for the other modeled factors.

Registered demand was also concentrated around morning and evening travel periods. A pronounced morning peak occurred around 08:00, while the highest average demand occurred at 17:00. Bicycle availability should therefore be protected before these peak periods, and operational activity that temporarily reduces available capacity should be scheduled away from them where practical.

The registered-user model had the largest daily error-persistence parameter, at 0.517. This indicates that unexplained deviations in registered demand showed greater adjacent-day persistence than those in the other two models. The result may provide useful context for short-term monitoring, but the model was not validated for forecasting.

Because the available data are aggregated at the system level, these findings identify when registered-user service reliability is most important but do not determine which individual stations require bicycles or empty docks. Station-level implementation would require inventory, origin–destination, and station-availability data.

### 6.2 Casual-User Operations

Casual users represented a smaller share of total demand than registered users, but their demand was more concentrated around weekends, holidays, and daytime activity. Casual demand therefore requires a different time-of-day operating profile from registered demand.

Average casual demand reached approximately 1,402 rentals per day on weekends, compared with approximately 625 rentals from Monday through Friday. Weekend casual demand was therefore 124.3% higher than weekday demand. The preferred casual-user model also retained separate weekday and holiday effects, and adjusted casual demand was approximately 396 rentals higher on holidays than on non-holidays.

The hourly profile was broader than the registered-user pattern. Casual demand increased gradually through the morning, remained elevated during the middle of the day, and reached its highest average level at 14:00. Operational preparation should therefore occur before the midday increase, while activities that temporarily reduce available capacity should be limited during the daytime and afternoon period where practical.

The casual-user model had the largest weekly error-persistence parameter, at 0.252. This indicates that unexplained deviations in casual demand retained more weekly structure than those in the other models. The parameter describes residual dependence rather than a direct effect of the previous week's rental volume.

The descriptive and calendar results support the use of separate operating templates for weekdays, weekends, and known holidays rather than one uniform weekly schedule. Weekend and holiday plans should also be adjusted for weather and other changing conditions. In addition, the available system-level data do not identify which locations attract the highest casual demand. Station-level allocation would require trip origin–destination records, station inventory, availability, and potentially event-related information.

### **6.3 Weather-Sensitive Planning**

Weather should adjust short-term operating plans, not determine them on its own. Descriptive results showed average daily demand of 4,877 rentals under clear or partly cloudy conditions, 4,036 under misty or cloudy conditions, and 1,803 under light rain or snow. Demand under light rain or snow was therefore approximately 63.0% lower than under clear or partly cloudy conditions.

The preferred models showed that the weather relationship remained substantial after adjustment for year, season, calendar structure, temperature, humidity, windspeed, and temporal dependence. The estimated adjusted difference between clear or partly cloudy conditions and light rain or snow was approximately 1,901 total rentals per day. The corresponding differences were approximately 333 casual rentals and 1,591 registered rentals.

Short-term operating plans should therefore be scaled according to expected weather conditions. Favorable-weather days require preparation for higher system activity, while light-rain or snow conditions may justify a lower-demand rebalancing profile. Any reduction in operational intensity should still preserve minimum service reliability, particularly for registered users, who continued to account for most system demand.

Weather conditions should also be considered together with calendar and user-segment patterns. For example, favorable weather on weekends may strengthen the already elevated casual-user demand, while adverse weather on working days does not remove the need to protect registered-user availability around morning and evening peaks. Weather-sensitive planning should therefore modify the relevant day-type operating plan rather than replace it.

These implications should be applied cautiously. Light rain or snow was observed on only 21 days, and the dataset contained no daily observations in the most severe weather category. The analysis also established adjusted associations rather than causal effects or validated forecasts. The results support flexible weather-based adjustments, but not fixed thresholds or automatic capacity changes.

### **6.4 Timing Implications for Fleet Operations**

Operating preparation should follow recurring demand profiles rather than one schedule applied uniformly across all days. Demand patterns change across months, day types, user segments, hours, and weather conditions.

Working days require particular readiness for registered-user demand. Registered users accounted for 86.8% of working-day rentals, and total hourly demand on working days reached its highest average level at 17:00. Capacity-reducing operational activity should therefore be completed before the evening peak where practical, while sufficient system availability should also be protected around the morning increase in registered demand.

Non-working days require a different timing profile. Total hourly demand on these days peaked earlier, at 13:00, and casual users represented 31.7% of daily rentals, compared with 13.2% on working days. Rebalancing resources should therefore be prepared earlier and should support the broader midday and afternoon demand period rather than focusing primarily on commuter peaks.

Longer-term fleet planning should also account for the variation observed across months and years. Demand in 2012 exceeded the corresponding 2011 level in every month, while activity rose during the warmer and higher-demand portions of the year before declining toward year-end. Periods of lower observed demand may provide more suitable windows for planned work that temporarily reduces available capacity, subject to minimum service requirements.

Weather should be used as an adjustment to these calendar and hourly operating profiles. Favorable conditions may require preparation for higher activity, whereas light rain or snow was associated with materially lower demand. Weather should not be used as the sole basis for reducing capacity because the adverse-weather sample was limited and registered-user service requirements remained operationally important.

The analysis identifies the periods of highest system-wide demand, but it cannot show which stations need bicycles or empty docks. Station-level allocation and rebalancing routes would require station inventory, dock availability, trip origin–destination, travel-time, operating-cost, and service-level data.

## 7 Recommendations

### 7.1 Priority 1 — Prioritize Working-Day Peak Readiness

Management should prioritize preparation for the main working-day demand periods. Registered users accounted for 86.8% of working-day rentals, and the preferred registered-user model estimated approximately 835 additional rentals on working days compared with non-working days after adjustment for the other modeled factors.

Within working days, registered-user demand showed two pronounced peaks. Average hourly registered demand reached approximately 455 rentals at 08:00 and its highest level, approximately 468 rentals, at 17:00. These patterns show that the system recorded its highest registered-user rental volumes around the morning and evening travel periods.

Planned system-wide activities that could temporarily reduce operating readiness should therefore be scheduled outside these peak periods where practical. Operational preparation should be completed before the morning and evening increases rather than beginning after demand has already reached its highest levels.

The available data measure completed rentals rather than bicycle availability, empty-dock availability, service interruptions, or unmet demand. These periods deserve greater operational attention, but the data do not show that service failures occurred or how much capacity was required.

Once station-level operational data become available, management should define and monitor working-day reliability indicators around the two peak periods. Relevant measures could include bicycle-availability rates, empty-dock availability, stockout duration, service interruptions, and estimated unmet demand.

## **7.2 Priority 2 — Use Segment- and Day-Type-Specific Operating Windows**

A single hourly operating schedule should not be applied uniformly across all days and user segments. The hourly analysis showed that the timing and composition of demand changed substantially between registered and casual users and between working and non-working days.

Across all dates, casual-user demand reached its highest average level at 14:00, whereas registered-user and total demand both peaked at 17:00. The day-type analysis provided additional operational detail. On working days, total and registered-user demand reached their highest average levels at 17:00, with a pronounced registered-user increase also visible around 08:00. On non-working days, total demand peaked earlier, at 13:00; casual demand peaked at 14:00, while registered demand reached its highest average level at 12:00.

Management should therefore use at least two distinct operating templates. The working-day template should prioritize preparation before the morning increase and the 17:00 evening peak. The non-working-day template should shift preparation toward the broader midday and early-afternoon period. Activities that could temporarily reduce system-wide operating readiness should be scheduled outside the relevant high-demand windows where practical.

The overall peak hours for casual, registered, and total demand remained unchanged when the analysis was restricted to dates with complete hourly coverage. This sensitivity result reduces concern that the main segment-level peak times were created by the missing date-hour combinations. Day-type-specific timings should nevertheless be monitored prospectively because a separate complete-day sensitivity test was not used to validate every day-type and segment combination.

The recommendation concerns timing only. The data cannot determine how many bicycles to move, which stations are involved, or which routes are most efficient. Those decisions require station-level inventory, bicycle and dock availability, origin–destination, travel-time, and operating-cost data.

When station-level data become available, management can test these operating windows in practice.. Performance should be compared across the working-day morning, working-day evening, and non-working-day midday windows using measures such as bicycle availability, empty-dock availability, stockout duration, completed rentals, and estimated unmet demand.

## **7.3 Priority 3 — Use a Weekend and Holiday Casual-Demand Readiness Plan**

Casual-user operations should use a distinct weekend and holiday readiness plan rather than relying on the working-day operating profile. Average casual demand was approximately 1,402 rentals per day on Saturdays and Sundays, compared with approximately 625 rentals from Monday through Friday. Weekend casual demand was therefore 124.3% higher than the weekday average.

The preferred casual-user model supported this pattern after adjustment for year, season, weather, temperature, humidity, windspeed, and temporal dependence. Saturday and Sunday showed the

strongest positive weekday deviations from the adjusted grand mean. Holidays were also associated with approximately 396 additional casual rentals per day compared with non-holidays.

The timing of operational preparation should also change. On non-working days, casual demand increased through the morning and reached its highest average level at 14:00. Operational preparation should therefore be completed before the midday increase, and planned activities that could temporarily reduce system-wide readiness should be scheduled outside the main midday and afternoon demand period where practical.

Weekend and holiday plans should remain weather-sensitive. Casual demand was positively associated with favorable weather and negatively associated with higher humidity, windspeed, and adverse weather conditions. The weekend operating template should therefore be adjusted using current weather information rather than applied as a fixed capacity rule.

These findings support prioritizing weekend and holiday readiness, but they do not establish that additional fleet capacity is required or quantify how many bicycles should be deployed. The data record completed rentals and do not include bicycle availability, empty-dock availability, stockouts, rejected trips, or unmet demand.

Once station-level operational data are available, management should test the weekend and holiday template using indicators such as bicycle availability, empty-dock availability, stockout duration, completed casual rentals, and estimated unmet demand. The results should then be used to determine whether additional fleet deployment or targeted rebalancing is operationally justified.

## **7.4 Priority 4 — Use Weather as a Short-Term Planning Adjustment**

Weather information should be used to adjust short-term operating plans rather than to determine capacity decisions independently. Historical average daily demand was 4,877 rentals under clear or partly cloudy conditions, 4,036 under misty or cloudy conditions, and 1,803 under light rain or snow. The observed average under light rain or snow was therefore approximately 63.0% lower than under clear or partly cloudy conditions.

The preferred models indicated that this relationship remained substantial after adjustment for year, season, calendar structure, temperature, humidity, windspeed, and temporal dependence. The estimated adjusted difference between clear or partly cloudy conditions and light rain or snow was approximately 1,901 total rentals per day. The corresponding adjusted differences were approximately 333 casual rentals and 1,591 registered rentals.

Management should first define the operating plan appropriate for the relevant day type and user-segment pattern, and then use available short-term weather information as an adjustment. Favorable conditions should trigger preparation for the higher-demand profile, while light rain or snow may support a lower-demand operating profile. Weather should not replace the working-day, non-working-day, or segment-specific planning templates established in the preceding recommendations.

The observed 63.0% demand difference should not be translated directly into a 63.0% reduction in fleet availability or operational effort. The dataset measures completed rentals rather than required fleet capacity, bicycle and dock availability, rejected trips, service failures, or unmet demand. Reducing readiness by the same percentage could therefore create unsupported operational risk.

The evidence for light rain or snow should also be applied cautiously because this category contained only 21 daily observations. No daily records were available for the most severe weather category, so the analysis does not support operational recommendations for heavy rain, severe snow, or fog conditions.

Management should first test weather-based adjustments rather than applying a fixed automatic threshold. For each operating day, the system should record the available weather information, the selected operating profile, completed rentals, bicycle and dock availability, stockout duration, service interruptions, and estimated unmet demand. These records can then show whether weather-specific thresholds are useful and how large the adjustments should be.

## **7.5 Priority 5 — Establish Segment-Level Demand Monitoring Before Long-Term Capacity Decisions**

Management should monitor casual and registered demand separately before making long-term fleet or capacity decisions. Average daily demand was 64.4% higher in 2012 than in 2011, and registered users accounted for 84.5% of the observed increase. Demand in 2012 also exceeded the corresponding 2011 average in all 12 calendar months.

The preferred models supported a substantial year difference after adjustment for season, calendar structure, weather, environmental conditions, and temporal dependence. Estimated adjusted daily demand in 2012 was approximately 2,008 total rentals higher than in 2011, including approximately 1,712 additional registered rentals and 277 additional casual rentals.

These results show that the system operated at a materially higher demand level in 2012, with most of the difference concentrated among registered users. However, the dataset covers only two years. It does not establish whether the increase continued after 2012, whether the rate of change was sustainable, or whether the difference resulted from user growth, service expansion, pricing, marketing, infrastructure changes, or other unobserved factors.

Management should therefore maintain a recurring segment-level monitoring process rather than extrapolating the observed two-year difference directly into future capacity requirements. The core monitoring set should include average daily rentals by segment, monthly year-over-year change, casual and registered shares of total demand, peak-hour timing by segment and day type, and demand by observed weather condition.

Monitoring should distinguish changes in demand volume from changes in user composition. A stable total-demand level may still conceal a meaningful shift between casual and registered users, as demonstrated by the working-day and non-working-day analysis. Segment shares should therefore be reviewed alongside total rentals rather than treated as secondary indicators.

The current rental data can support demand monitoring but cannot determine whether capacity is sufficient. Before committing to fleet expansion or other major operational investment, management should supplement rental counts with station-level bicycle and dock availability, stockout duration, rejected or abandoned trips, service interruptions, operating costs, and estimated unmet demand.

Long-term decisions should then be based on repeated evidence across multiple periods rather than a single year-to-year comparison. The 2011–2012 results provide a useful historical baseline, but they do not forecast continued growth.



Table 4. Management action framework

| Priority                                | Evidence basis                                                                                                           | Recommended management action                                                                            | Main validation requirement                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Working-day peak readiness              | Registered users represented 86.8% of working-day demand; strongest peaks occurred around 08:00 and 17:00                | Complete operational preparation before the morning and evening demand increases                         | Station availability, stockouts, service interruptions, and unmet demand             |
| Segment- and day-type operating windows | Casual demand peaked at 14:00; registered and total demand peaked at 17:00; non-working-day total demand peaked at 13:00 | Use separate working-day and non-working-day operating templates                                         | Prospective comparison of availability and service outcomes across operating windows |
| Weekend and holiday casual readiness    | Weekend casual demand was 124.3% higher than the weekday average; holidays had a positive adjusted association           | Prepare the casual-user operating profile before the midday and afternoon demand period                  | Station-level casual demand, availability, stockouts, and unmet demand               |
| Weather adjustment                      | Light rain or snow was associated with substantially lower demand, but only 21 such days were observed                   | Adjust the relevant day-type plan using short-term weather information; avoid fixed automatic reductions | Prospective weather, availability, service, and unmet-demand records                 |
| Segment-level monitoring                | Average daily demand was 64.4% higher in 2012; registered users contributed 84.5% of the increase                        | Track demand volume and user composition before long-term capacity commitments                           | Multiple years of demand and operational-capacity evidence                           |

*Note:* Recommended actions identify planning priorities supported by the observed demand patterns. They do not specify station-level fleet quantities, rebalancing routes, or investment requirements because the available data contain completed rentals rather than direct measures of capacity adequacy or unmet demand.

*Source:* Author's synthesis of the validated descriptive and preferred-model results.

## 8 Risks and Limitations

The findings and recommendations in this report should be interpreted within the following limitations.

1. **Observational design and causal interpretation.** The analysis is based on observational records and does not use randomization, a natural experiment, or a formal causal-identification strategy. Descriptive differences and model coefficients therefore represent observed or adjusted associations. They do not establish that calendar status, weather, temperature, humidity, windspeed, or any other modeled variable caused the corresponding change in rental demand.
2. **Limited time horizon and generalizability.** The data cover January 2011 through December 2012 only. The higher demand observed in 2012 provides a valid historical comparison but



does not establish a continuing long-term growth trend. The findings may not generalize to later operating conditions, particularly if the system subsequently changed its service area, fleet size, membership base, infrastructure, pricing, promotional activity, or operating policies.

3. System-level rather than station-level scope. The available records contain aggregated completed rentals and do not include station inventory, bicycle availability, empty-dock availability, trip origins and destinations, rejected or abandoned trips, service interruptions, rebalancing movements, travel times, or operating costs. The analysis can identify periods of higher system-wide rental activity, but it cannot determine whether capacity was sufficient, which stations required intervention, how many bicycles should be moved, or whether a proposed action would be financially justified.
4. Incomplete hourly coverage and environmental measurement uncertainty. The hourly dataset covers all 731 dates, but 165 expected date–hour combinations are absent across 76 dates. The overall casual, registered, and total peak hours remained unchanged when the analysis was restricted to the 655 dates with complete hourly coverage, reducing concern that the principal peak-hour findings were driven by the missing records. Nevertheless, incomplete coverage may still affect more detailed estimates for particular day-type and user-segment combinations.  
Humidity was also recorded as zero on March 10, 2011, in one daily record and 22 hourly records. The original values were retained and flagged because the available data did not provide sufficient evidence for defensible corrected values.
5. Uneven weather coverage. Light rain or snow was observed on only 21 days, and no daily observations were available for the most severe weather category. Estimated relationships for adverse weather should therefore be applied cautiously and should not be generalized to heavy rain, severe snow, or fog. Observed reductions in completed rentals should also not be converted directly into equivalent reductions in fleet availability or operational readiness.
6. Model specification and remaining temporal dependence. The preferred models use linear fixed effects together with daily and weekly autoregressive error components. Their coefficients summarize average adjusted relationships and may not capture nonlinear responses, interactions, unusual events, or structural changes.  
The selected temporal structures substantially reduced residual correlation at the one-day and seven-day lags. However, at least one Ljung–Box test indicated remaining serial dependence for each response, with the clearest remaining dependence in the casual-user model. The preferred models should therefore not be presented as fully specified or independently validated forecasting systems.
7. Validation scope and variable interpretation. Cross-tool agreement between JMP and Python was established for the total-demand OLS baseline. The preferred models with daily and weekly autoregressive structure were subsequently estimated and diagnosed in Python rather than independently reproduced in JMP. Model selection considered convergence, AIC and BIC within each response, residual diagnostics, simplicity, and operational

interpretability. The selection process was not intended to identify the most complex possible specification. Windspeed is reported only on its retained original scale because its physical unit was not independently verified.

These limitations do not by themselves invalidate the main descriptive patterns or adjusted relationships. Instead, they define the appropriate scope of interpretation. The findings support system-level planning priorities and the design of monitored operational tests, but they do not support causal claims, automatic capacity rules, station-level optimization, or unvalidated long-term forecasts.

## 9 Methodology Summary

The project followed a staged and reproducible analytical workflow covering data audit, data preparation, descriptive analysis, statistical modeling, diagnostic evaluation, and explanatory reporting.

The raw daily and hourly datasets were audited separately before any transformations were applied. The audit reviewed dataset structure, observation counts, missing values, duplicate records, categorical ranges, rental-count identities, date and calendar consistency, hourly-record completeness, reconciliation of daily totals with aggregated hourly records, environmental-variable boundaries, and consistency between the two datasets. The original source files were not modified. Identified data-quality issues were recorded in a formal issue register and retained for sensitivity analysis and reporting.

Cleaning and preparation were performed on separate working copies. Dates were converted to structured date values, hourly timestamps were created, documented data-quality flags were added, environmental variables and category labels were prepared for interpretation, and records were sorted chronologically. No rental observations were deleted or imputed. The original observation counts and rental totals were preserved, and the validated daily and hourly datasets were exported for subsequent analysis.

Exploratory analysis was conducted separately at the daily and hourly levels. Daily analysis examined overall demand, annual and monthly differences, seasonal and calendar patterns, user composition, and weather-related variation. Hourly analysis examined within-day demand and segment-specific peak timing. Casual, registered, and total rentals were evaluated separately where operationally relevant. Weather-category sample sizes were reported, and continuous weather variables were treated as associated factors rather than isolated causal drivers.

Two documented data-quality concerns were evaluated through sensitivity analysis. The zero-humidity observation was excluded temporarily to assess its influence on weather correlations, while hourly peak analyses were repeated using only dates with complete 24-hour coverage. The zero-humidity record had only a limited effect on the examined correlations, and the principal casual, registered, and total peak hours remained unchanged in the complete-date analysis. The original observations were therefore retained and flagged rather than replaced.

Statistical analysis began in JMP with distribution reviews, bivariate and multivariate exploration, ordinary least squares regression, and residual diagnostics. The total-demand OLS baseline was then reproduced in Python using the same predictors and sum-coding structure. Agreement in coefficients, model fit, and principal diagnostics provided a cross-tool validation check for that baseline specification.

Because the OLS residuals showed positive temporal autocorrelation, the analysis was extended in Python using regression models with autoregressive errors. Candidate specifications included the OLS baseline, a daily AR(1) structure, and a combined daily AR(1) and weekly seasonal AR(1) structure with a seven-day period. Models were estimated separately for total, casual, and registered demand. The preferred casual-user model used weekday and holiday effects, while the total-demand and registered-user models used working-day status.

Preferred models were selected separately for each response using convergence, AIC and BIC comparisons within the same response, residual autocorrelation, Durbin–Watson and Ljung–Box diagnostics, model simplicity, and operational interpretability. Coefficients, standard errors, confidence intervals, p-values, temporal parameters, and residual diagnostics were exported as structured reporting tables.

The final communication stage used validated analytical outputs to create six explanatory figures. Each figure was designed around one management message, with explicit units, restrained visual design, and interpretation limitations. Descriptive evidence, adjusted model results, operational implications, recommendations, and limitations were kept distinct throughout the report. The workflow was designed to support reproducibility and defensible system-level interpretation; it was not designed to establish causality, optimize individual stations, or produce a fully validated forecasting system.

## 10 Conclusion

The analysis shows that bike-sharing rental activity differed materially across years, user segments, calendar conditions, hours of the day, and weather categories. Average daily rentals were 64.4% higher in 2012 than in 2011, and registered users accounted for 84.5% of the observed increase. The higher 2012 level was visible across all 12 calendar months, although the two-year period is not sufficient to establish a continuing long-term growth trend.

Total rental volume alone did not fully describe the system’s operating patterns. Registered users dominated working-day activity and recorded their highest hourly rental volumes around the morning and evening travel periods, with the overall registered-user peak occurring at 17:00. Casual-user activity followed a broader daytime profile, peaked at 14:00, and was substantially higher on weekends than from Monday through Friday. Working and non-working days therefore require different planning templates even when their average total rental volumes appear relatively similar.

Weather was also associated with meaningful variation in completed rentals. Light rain or snow was associated with substantially lower observed and model-adjusted rental activity than clear or partly cloudy conditions. However, the limited number of adverse-weather observations means that weather should be used to adjust an existing day-type and segment-specific plan rather than to trigger fixed or automatic capacity reductions.

The preferred time-series regression models generally reinforced the principal descriptive findings after adjustment for year, season, calendar structure, weather, temperature, humidity, windspeed, and temporal dependence. The models substantially reduced immediate residual autocorrelation, but some serial dependence remained, particularly for casual-user demand. Their results should therefore be interpreted as adjusted associations for planning support rather than causal estimates or fully validated forecasts.

Management should test separate operating plans for working and non-working days, for casual and registered users, and for different weather conditions. Operational preparation should receive greater attention before the registered-user morning and evening peaks and before the casual-user midday and afternoon period on weekends and holidays. Segment-level monitoring should also continue before any long-term fleet or capacity commitment is made.

The analysis identifies the periods when system-wide demand requires the most attention, but it cannot determine whether capacity was sufficient, which stations required intervention, or how many bicycles should be moved. Station-level inventory, bicycle and dock availability, origin–destination records, service interruptions, operating costs, and unmet-demand measures are needed for those decisions.

Overall, the project provides a validated and reproducible historical baseline for system-level planning. Rather than offering one operating rule for all conditions, the project provides a basis for testing different plans across user segments, day types, and weather conditions.

## Appendix A — Data-Quality Issue Register

The data audit identified two issues that required formal documentation and sensitivity assessment. Neither issue was corrected automatically because the available records did not provide sufficient evidence for defensible replacement values.

Both issues were retained in the analytical datasets, flagged where appropriate, and considered during interpretation. Table 5 summarizes the issue register, while the following subsections describe the evidence, treatment, validation checks, and remaining limitations associated with each issue.

Table 5. Data-quality issue register

| Issue ID | Dataset          | Identified issue                                                                    | Severity | Analytical treatment                                                                                                                         |
|----------|------------------|-------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| DQ-01    | Hourly           | 165 expected date–hour combinations were absent across 76 dates                     | Medium   | Original records retained; absent records were not treated as zero demand; peak-hour sensitivity analysis was performed using complete dates |
| DQ-02    | Daily and hourly | Humidity was recorded as zero on March 10, 2011, in one daily and 22 hourly records | Medium   | Original values retained and flagged; influence on weather-related results was evaluated through sensitivity analysis                        |

*Note:* Severity reflects the data-quality classification used in the project issue register. It does not represent an estimate of operational or financial impact.

*Source:* Author’s structured data audit and issue register.

### Issue DQ-01 — Missing Date–Hour Combinations

The hourly-data audit identified 165 expected date–hour combinations that were absent across 76 of the 731 dates in the analysis period. Every calendar date was represented in the hourly dataset, but only 655 dates contained the complete set of 24 hourly records.

The absent combinations were treated as gaps in hourly coverage rather than observations of zero demand. No records were created and no rental counts were imputed because the source data did not provide sufficient evidence for estimating the unobserved hourly values. The original hourly dataset was therefore retained, and the affected dates were documented separately.

Cross-dataset reconciliation showed that the sum of the available hourly rental records matched the corresponding daily rental total for every date. The identified issue therefore concerns the completeness of within-day timing information rather than a disagreement between the daily and hourly rental totals.

To evaluate the effect on the principal hourly findings, the peak-hour analysis was repeated using only the 655 dates with complete 24-hour coverage. The results were unchanged: casual-user demand continued to peak at 14:00, while registered-user and total demand continued to peak at 17:00.

This sensitivity result increases confidence that the main overall peak-hour findings were not created by the missing date-hour combinations. It does not demonstrate that every detailed estimate by day type, user segment, or individual hour is unaffected. More granular hourly comparisons should therefore retain the documented coverage limitation.

## **Issue DQ-02 — Zero Humidity Records**

The data audit identified a humidity value of zero on March 10, 2011. The same value appeared in the daily record for that date and in all 22 available hourly records. Because the value was repeated consistently across the daily and hourly datasets, the issue was not treated as a simple transcription mismatch between the two files.

The available source records did not provide an independently verified humidity value for that date. Replacing the recorded value with a mean, median, neighboring-day value, or model-based estimate would therefore introduce an unsupported assumption. The original values were retained, and the affected daily and hourly records were flagged for traceability.

A sensitivity analysis was conducted by temporarily excluding the affected daily observation from the examined weather-correlation calculations. The comparison showed only limited changes in the correlations and did not alter the main interpretation of the weather-related relationships. The record was therefore retained in the final descriptive and regression analyses rather than deleted or imputed.

This sensitivity check does not establish that the recorded zero is correct, nor does it prove that every model coefficient is completely unaffected. It shows only that the principal weather-related interpretation examined in the project was not materially dependent on this single daily observation.

The issue should remain documented because an extreme environmental value can influence estimated relationships, particularly in a relatively short dataset. If verified historical weather records become available, the March 10, 2011 value should be compared with an external source before any correction is considered.

## Appendix B — Preferred Time-Series Models

Three response-specific time-series regression models were selected for total, casual-user, and registered-user daily rentals. All models were estimated using the 731 daily observations covering January 2011 through December 2012.

The models shared a common set of year, season, weather, temperature, humidity, and windspeed predictors. Their calendar specifications differed according to the observed demand structure. Total and registered-user demand were modeled using working-day status, whereas casual-user demand was modeled using weekday and holiday effects.

Each preferred specification included a first-order daily autoregressive error component and a first-order seasonal autoregressive error component with a seven-day period. This structure was selected to represent persistence from one day to the next and recurring weekly dependence that remained after accounting for the observed predictors.

The models were selected separately for each response. Their AIC and BIC values are therefore relevant only when comparing candidate specifications for the same dependent variable. Model coefficients represent adjusted associations conditional on the selected specification and should not be interpreted as causal effects.

This appendix documents the exact fixed-effect specifications, coding conventions, parameter estimates, temporal structures, and residual diagnostics used in the final report.

### B.1 Specification and Coding Conventions

The total-demand and registered-user models used the following fixed-effect specification:

```
response ~ C(yr, Sum) + C(season_label, Sum)
          + C(workingday, Sum)
          + C(weather_label, Sum)
          + temp_c + humidity_pct
          + windspeed_original_scale
```

The casual-user model used the following fixed-effect specification:

```
casual ~ C(yr, Sum) + C(season_label, Sum)
        + C(weekday, Sum) + C(holiday, Sum)
        + C(weather_label, Sum)
        + temp_c + humidity_pct
        + windspeed_original_scale
```

No interaction terms or nonlinear transformations were included in the preferred fixed-effect specifications. The continuous coefficients therefore represent the estimated average change in daily rentals associated with a one-unit increase in the relevant predictor, holding the other included variables constant. The temperature coefficient corresponds to a one-degree Celsius increase, the humidity coefficient to a one-percentage-point increase, and windspeed is interpreted only on its retained original scale.

Categorical predictors were represented using sum coding rather than reference-category coding. Under this coding system, the displayed coefficient for a category is a deviation from the model intercept while the other predictors are held constant. The intercept should not be interpreted as the raw sample mean because it is also conditional on the continuous predictors being equal to zero.

For a binary sum-coded factor, the difference between the two categories equals twice the magnitude of the displayed deviation coefficient, with the sign determined by the direction of comparison. For factors with more than two categories, the coefficient of the non-displayed category is the negative sum of the displayed category coefficients. Differences between individual categories are calculated by subtracting their corresponding deviations.

The non-displayed categories were 2012 for year, winter for season, mist or cloudy conditions for weather, working day for working-day status, Saturday for weekday, and holiday for holiday status. These are inferred sum-coded levels rather than conventional reference categories.

The error term in every preferred model included a daily AR(1) component and a seasonal AR(1) component at lag seven. These parameters describe dependence remaining in the model errors after adjustment for the fixed effects. They should not be interpreted as direct effects of the previous day's or previous week's rental volume.

Model-reported standard errors, confidence intervals, and p-values are conditional on the selected specification and its assumptions. Statistical significance was not treated as sufficient evidence of causal importance or operational value. Interpretation also considered effect magnitude, descriptive evidence, temporal diagnostics, data coverage, and the practical limits of the available system-level rental records.

## **B.2 Total-Demand Preferred Model**

The preferred total-demand model used daily total completed rentals (cnt) as the response variable and the working-day fixed-effect specification defined in Section B.1. The model was estimated using all 731 daily observations and included a daily AR(1) error component together with a weekly seasonal AR(1) component at a seven-day lag.

The model converged successfully. Its AIC was 11,718.59 and its BIC was 11,782.92. These information criteria describe the selected total-demand specification and should be used only when comparing candidate models fitted to the same total-demand response.

Table 6 reports the complete fixed-effect, temporal-structure, and variance parameter estimates. Because categorical predictors were sum coded, the displayed categorical coefficients represent deviations associated with the displayed levels rather than conventional reference-category comparisons. The intercept is conditional on the continuous predictors being equal to zero and should not be interpreted as the raw average daily rental count.

Table 6. Preferred total-demand model parameter estimates

| Parameter type | Parameter                        | Estimate   | Standard error | 95% confidence interval  | p-value |
|----------------|----------------------------------|------------|----------------|--------------------------|---------|
| Fixed effect   | Intercept                        | 3,612.16   | 249.80         | [3,122.56, 4,101.77]     | <0.001  |
| Fixed effect   | 2011 deviation                   | -1,004.06  | 59.01          | [-1,119.72, -888.40]     | <0.001  |
| Fixed effect   | Spring deviation                 | -888.74    | 120.04         | [-1,124.01, -653.47]     | <0.001  |
| Fixed effect   | Summer deviation                 | 260.40     | 90.54          | [82.93, 437.86]          | 0.004   |
| Fixed effect   | Fall deviation                   | -26.65     | 106.68         | [-235.74, 182.45]        | 0.803   |
| Fixed effect   | Non-working-day deviation        | -81.11     | 30.57          | [-141.02, -21.20]        | 0.008   |
| Fixed effect   | Clear or partly cloudy deviation | 777.52     | 61.63          | [656.73, 898.30]         | <0.001  |
| Fixed effect   | Light rain or snow deviation     | -1,123.59  | 77.67          | [-1,275.81, -971.37]     | <0.001  |
| Fixed effect   | Temperature, per 1°C             | 111.89     | 7.50           | [97.18, 126.59]          | <0.001  |
| Fixed effect   | Humidity, per percentage point   | -16.59     | 2.82           | [-22.11, -11.07]         | <0.001  |
| Fixed effect   | Windspeed, original scale        | -31.46     | 5.24           | [-41.73, -21.20]         | <0.001  |
| Time structure | Daily AR(1)                      | 0.4611     | 0.0246         | [0.4130, 0.5093]         | <0.001  |
| Time structure | Weekly seasonal AR(1), lag 7     | 0.1109     | 0.0327         | [0.0469, 0.1750]         | <0.001  |
| Variance       | Residual variance, $\sigma^2$    | 517,434.77 | 20,039.92      | [478,157.24, 556,712.29] | <0.001  |

Note: Categorical coefficients use sum coding and represent deviations associated with the displayed categories. The non-displayed deviations are 1,004.06 for 2012, 654.99 for winter, 81.11 for working days, and 346.07 for misty or cloudy conditions. Continuous-variable estimates represent adjusted average associations per one-unit increase. Windspeed is interpreted only on its retained original scale. The AR parameters describe dependence remaining in the model errors after adjustment for the fixed effects.

Source: Author's preferred-model outputs from 05\_all\_preferred\_model\_parameters.csv and 01\_preferred\_model\_summary.csv.

## B.2.1 Interpretation of Selected Adjusted Relationships

The year coefficient implies an adjusted difference of approximately 2,008 total rentals per day between 2012 and 2011. This contrast is obtained from the two sum-coded year deviations and indicates that adjusted total rental activity was higher in 2012. It describes an association within the observed two-year period and does not establish a continuing growth trend.

The working-day coefficient implies approximately 162 additional total rentals on working days compared with non-working days, after adjustment for the other modeled variables and temporal dependence. This adjusted difference is smaller than the substantial change in user composition between the two day types and should not be interpreted as evidence that their operational requirements were equivalent.

The estimated adjusted difference between clear or partly cloudy conditions and light rain or snow was approximately 1,901 total rentals per day. Temperature was associated with approximately 112 additional daily rentals per one-degree Celsius increase, while humidity was associated with approximately 17 fewer rentals per percentage-point increase. Windspeed was also negatively associated with total rentals, but its coefficient should be interpreted only on the retained original scale.



The seasonal coefficients are deviations under sum coding rather than direct comparisons with a reference season. Spring had a substantial negative deviation, summer had a positive deviation, and the fall deviation was small and not statistically distinguishable from zero under the selected specification. Any direct seasonal comparison must be calculated by subtracting the relevant sum-coded deviations.

### **B.2.2 Temporal Structure and Residual Diagnostics**

The estimated daily AR(1) parameter was 0.4611, indicating positive dependence between adjacent-day disturbances in the fitted error process. The weekly seasonal AR(1) parameter was 0.1109, indicating a smaller but statistically supported seven-day error component. These parameters characterize residual temporal dependence and are not coefficients on observed prior-day rental counts.

After inclusion of the daily and weekly autoregressive components, residual autocorrelation was  $-0.0428$  at lag 1 and  $-0.0101$  at lag 7. The Durbin–Watson statistic was 2.0847, indicating that the strong positive lag-1 autocorrelation observed in the OLS baseline was substantially reduced.

The Ljung–Box p-values were 0.0117 at lag 7, 0.0011 at lag 14, and 0.0057 at lag 30. These joint tests indicate that some residual serial dependence remained despite the improvement at the individual one-day and seven-day lags. The preferred model therefore provides a stronger inferential specification than the OLS baseline, but it should not be described as a fully specified or independently validated forecasting model.

## **B.3 Casual-User Preferred Model**

The preferred casual-user model used daily casual rentals as the response variable and the weekday-and-holiday fixed-effect specification defined in Section B.1. The model was estimated using all 731 daily observations and included a daily AR(1) error component together with a weekly seasonal AR(1) component at a seven-day lag.

Unlike the total-demand and registered-user models, the casual-user specification represented calendar structure through separate weekday and holiday terms rather than a single working-day indicator. This specification was retained because casual demand showed substantial variation across individual weekdays that would be concealed by a binary working-day classification.

The model converged successfully. Its AIC was 10,541.41 and its BIC was 10,633.30. These information criteria describe the selected casual-user specification and should be used only when comparing candidate models fitted to the same casual-demand response.

Table 7 reports the complete fixed-effect, temporal-structure, and variance parameter estimates. The displayed categorical coefficients use sum coding and therefore represent deviations

associated with the displayed levels. The intercept is conditional on the continuous predictors being equal to zero and should not be interpreted as the raw average number of daily casual rentals.

Table 7. Preferred casual-user model parameter estimates

| Parameter type | Parameter                        | Estimate   | Standard error | 95% confidence interval | p-value |
|----------------|----------------------------------|------------|----------------|-------------------------|---------|
| Fixed effect   | Intercept                        | 873.49     | 111.04         | [655.86, 1,091.13]      | <0.001  |
| Fixed effect   | 2011 deviation                   | -138.49    | 27.74          | [-192.85, -84.12]       | <0.001  |
| Fixed effect   | Spring deviation                 | -144.40    | 55.86          | [-253.88, -34.91]       | 0.010   |
| Fixed effect   | Summer deviation                 | 175.50     | 39.15          | [98.77, 252.23]         | <0.001  |
| Fixed effect   | Fall deviation                   | -45.02     | 52.68          | [-148.27, 58.24]        | 0.393   |
| Fixed effect   | Sunday deviation                 | 511.91     | 39.28          | [434.92, 588.90]        | <0.001  |
| Fixed effect   | Monday deviation                 | -221.93    | 50.72          | [-321.35, -122.52]      | <0.001  |
| Fixed effect   | Tuesday deviation                | -284.62    | 61.27          | [-404.71, -164.53]      | <0.001  |
| Fixed effect   | Wednesday deviation              | -288.16    | 58.84          | [-403.48, -172.84]      | <0.001  |
| Fixed effect   | Thursday deviation               | -275.99    | 68.41          | [-410.06, -141.91]      | <0.001  |
| Fixed effect   | Friday deviation                 | -102.20    | 46.32          | [-192.99, -11.41]       | 0.027   |
| Fixed effect   | Non-holiday deviation            | -198.22    | 28.55          | [-254.16, -142.27]      | <0.001  |
| Fixed effect   | Clear or partly cloudy deviation | 143.50     | 29.97          | [84.77, 202.23]         | <0.001  |
| Fixed effect   | Light rain or snow deviation     | -189.82    | 44.19          | [-276.42, -103.21]      | <0.001  |
| Fixed effect   | Temperature, per 1°C             | 39.53      | 3.62           | [32.43, 46.63]          | <0.001  |
| Fixed effect   | Humidity, per percentage point   | -5.65      | 1.26           | [-8.12, -3.18]          | <0.001  |
| Fixed effect   | Windspeed, original scale        | -14.79     | 2.35           | [-19.40, -10.18]        | <0.001  |
| Time structure | Daily AR(1)                      | 0.3676     | 0.0275         | [0.3137, 0.4214]        | <0.001  |
| Time structure | Weekly seasonal AR(1), lag 7     | 0.2523     | 0.0261         | [0.2011, 0.3036]        | <0.001  |
| Variance       | Residual variance, $\sigma^2$    | 101,392.08 | 3,837.52       | [93,870.66, 108,913.49] | <0.001  |

Note: Categorical coefficients use sum coding and represent deviations associated with the displayed categories. The inferred non-displayed deviations are 138.49 for 2012, 13.92 for winter, 660.99 for Saturday, 198.22 for holidays, and 46.32 for misty or cloudy conditions. The inferred coefficients do not have separate standard errors or p-values in this displayed parameterization. Continuous-variable estimates represent adjusted average associations per one-unit increase. Windspeed is interpreted only on its retained original scale. The AR parameters describe dependence remaining in the model errors after adjustment for the fixed effects.

Source: Author's preferred-model outputs from 05\_all\_preferred\_model\_parameters.csv and 01\_preferred\_model\_summary.csv.

### B.3.1 Interpretation of Selected Adjusted Relationships

The year coefficient implies an adjusted difference of approximately 277 casual rentals per day between 2012 and 2011. This result indicates that casual rental activity was higher in 2012 after

adjustment for the included calendar, weather, environmental, and temporal factors. It does not establish that the difference represented a continuing growth trend.

The weekday estimates show that casual demand followed a distinct weekly structure. Saturday had an inferred positive deviation of approximately 661 rentals and Sunday had a displayed positive deviation of approximately 512 rentals. The displayed Monday-through-Friday deviations were negative, although the Friday deviation was smaller in magnitude than those for Monday through Thursday. These estimates are consistent with the descriptive finding of substantially higher casual demand during weekends.

The holiday term implies an adjusted difference of approximately 396 casual rentals per day between holidays and non-holidays. Because the displayed coefficient represents the non-holiday deviation of  $-198.22$ , the inferred holiday deviation is  $198.22$  and the corresponding holiday-versus-non-holiday contrast is  $396.43$  rentals. This association should not be interpreted as a causal holiday effect.

The estimated adjusted difference between clear or partly cloudy conditions and light rain or snow was approximately 333 casual rentals per day. Temperature was associated with approximately 40 additional casual rentals per one-degree Celsius increase, while humidity was associated with approximately six fewer casual rentals per percentage-point increase. Windspeed was also negatively associated with casual demand on its retained original scale.

Spring had a negative adjusted deviation and summer had a positive adjusted deviation. The fall deviation was relatively small and was not statistically distinguishable from zero under the selected specification. Because the seasonal terms use sum coding, direct comparisons between seasons require subtraction of the relevant deviations.

### **B.3.2 Temporal Structure and Residual Diagnostics**

The estimated daily AR(1) parameter was 0.3676, while the weekly seasonal AR(1) parameter was 0.2523. Both parameters were positive and statistically supported, indicating that the selected error structure captured dependence at the adjacent-day and seven-day frequencies after adjustment for the fixed effects. The comparatively larger weekly parameter in the casual-user model is consistent with the pronounced weekly pattern in casual demand, but it should not be interpreted as a direct effect of the previous week's rental volume.

After inclusion of the autoregressive components, residual autocorrelation was 0.0124 at lag 1 and  $-0.0416$  at lag 7. The Durbin–Watson statistic was 1.9702. These diagnostics show that the individual one-day and seven-day residual correlations were reduced to values close to zero.

However, the Ljung–Box p-values were below 0.001 at lags 7, 14, and 30. These joint tests provide strong evidence that additional serial dependence remained across the examined sets of residual lags. The remaining dependence was more pronounced for casual demand than for the total-demand and registered-user models.

The selected model therefore improves substantially on the OLS baseline and provides a defensible adjusted-association model for explanatory reporting. It should nevertheless not be presented as a fully specified forecasting model, and its uncertainty estimates remain conditional on the selected temporal structure.

## B.4 Registered-User Preferred Model

The preferred registered-user model used daily registered rentals as the response variable and the working-day fixed-effect specification defined in Section B.1. The model was estimated using all 731 daily observations and included a daily AR(1) error component together with a weekly seasonal AR(1) component at a seven-day lag.

The working-day specification was retained because registered-user activity showed a strong calendar distinction between working and non-working days. The model estimates this difference while adjusting for year, season, weather category, temperature, humidity, windspeed, and remaining daily and weekly temporal dependence.

The model converged successfully. Its AIC was 11,318.95 and its BIC was 11,383.27. These information criteria describe the selected registered-user specification and should be used only when comparing candidate models fitted to the same registered-demand response.

Table 8 reports the complete fixed-effect, temporal-structure, and variance parameter estimates. The categorical predictors use sum coding, so the displayed coefficients represent deviations associated with the displayed levels rather than conventional comparisons with reference categories. The intercept is conditional on the continuous predictors being equal to zero and should not be interpreted as the raw average number of daily registered rentals.

Table 8. Preferred registered-user model parameter estimates

| Parameter type | Parameter                        | Estimate   | Standard error | 95% confidence interval  | p-value |
|----------------|----------------------------------|------------|----------------|--------------------------|---------|
| Fixed effect   | Intercept                        | 2,806.13   | 195.52         | [2,422.92, 3,189.35]     | <0.001  |
| Fixed effect   | 2011 deviation                   | -855.96    | 48.82          | [-951.65, -760.28]       | <0.001  |
| Fixed effect   | Spring deviation                 | -712.97    | 97.25          | [-903.57, -522.36]       | <0.001  |
| Fixed effect   | Summer deviation                 | 69.51      | 77.91          | [-83.18, 222.20]         | 0.372   |
| Fixed effect   | Fall deviation                   | 21.52      | 86.09          | [-147.21, 190.25]        | 0.803   |
| Fixed effect   | Non-working-day deviation        | -417.60    | 22.87          | [-462.42, -372.78]       | <0.001  |
| Fixed effect   | Clear or partly cloudy deviation | 646.56     | 45.79          | [556.82, 736.30]         | <0.001  |
| Fixed effect   | Light rain or snow deviation     | -943.96    | 55.71          | [-1,053.15, -834.76]     | <0.001  |
| Fixed effect   | Temperature, per 1°C             | 74.04      | 5.90           | [62.48, 85.61]           | <0.001  |
| Fixed effect   | Humidity, per percentage point   | -11.35     | 2.16           | [-15.59, -7.12]          | <0.001  |
| Fixed effect   | Windspeed, original scale        | -17.67     | 4.15           | [-25.81, -9.54]          | <0.001  |
| Time structure | Daily AR(1)                      | 0.5168     | 0.0227         | [0.4724, 0.5612]         | <0.001  |
| Time structure | Weekly seasonal AR(1), lag 7     | 0.1012     | 0.0384         | [0.0259, 0.1765]         | 0.008   |
| Variance       | Residual variance, $\sigma^2$    | 300,759.40 | 11,976.38      | [277,286.14, 324,232.67] | <0.001  |

*Note: Categorical coefficients use sum coding and represent deviations associated with the displayed categories. The inferred non-displayed deviations are 855.96 for 2012, 621.93 for winter, 417.60 for working days, and 297.40 for misty or cloudy conditions. These inferred coefficients do not have separate standard errors or p-values in the displayed parameterization; formal inference for a specific comparison requires an appropriate linear contrast.*

*Continuous-variable estimates represent adjusted average associations per one-unit increase. Windspeed is interpreted only on its retained original scale. The AR parameters describe dependence remaining in the model errors after adjustment for the fixed effects.*

*Source: Author's preferred-model outputs from 05\_all\_preferred\_model\_parameters.csv and 01\_preferred\_model\_summary.csv.*

#### **B.4.1 Interpretation of Selected Adjusted Relationships**

The year coefficient implies an adjusted difference of approximately 1,712 registered rentals per day between 2012 and 2011. This contrast is obtained from the two sum-coded year deviations and indicates that registered-user rental activity was substantially higher in 2012 after adjustment for the included predictors and temporal dependence. It describes a historical association within the observed period and does not establish continued growth beyond 2012.

The working-day coefficient implies an adjusted difference of approximately 835 registered rentals per day between working and non-working days. The displayed non-working-day deviation is -417.60, while the inferred working-day deviation is 417.60. This result supports the descriptive evidence that registered users were more strongly concentrated on working days. It does not establish that working-day status itself caused the difference.

The estimated adjusted difference between clear or partly cloudy conditions and light rain or snow was approximately 1,591 registered rentals per day. Temperature was associated with approximately 74 additional registered rentals per one-degree Celsius increase, whereas humidity was associated with approximately 11 fewer registered rentals per percentage-point increase. Windspeed was also negatively associated with registered demand on its retained original scale.

Spring had a substantial negative adjusted deviation. The summer and fall deviations were relatively small and were not statistically distinguishable from zero under the selected specification. Winter had an inferred positive deviation of approximately 622 registered rentals, but direct statistical inference for the inferred winter coefficient or for a specific seasonal comparison requires an appropriate linear contrast.

These estimates describe conditional average relationships within the selected model. Their magnitude should be interpreted together with the descriptive results, residual diagnostics, data coverage, and the absence of station-level availability and unmet-demand measures.

#### **B.4.2 Temporal Structure and Residual Diagnostics**

The estimated daily AR(1) parameter was 0.5168, indicating substantial positive dependence between adjacent-day model disturbances after adjustment for the fixed effects. The weekly seasonal AR(1) parameter was 0.1012, indicating a smaller but statistically supported seven-day component. These parameters characterize the correlation structure of the model errors and should not be interpreted as direct effects of observed rental demand on previous days.

After inclusion of the daily and weekly autoregressive components, residual autocorrelation was -0.0259 at lag 1 and -0.0122 at lag 7. The Durbin-Watson statistic was 2.0513. These results indicate that the strong positive short-term autocorrelation observed in the registered-user OLS residuals was substantially reduced.

The Ljung–Box p-value was 0.1532 at lag 7, providing no evidence of remaining joint serial dependence across the first seven lags at the 5% significance level. The corresponding p-value was 0.0169 at lag 14, indicating evidence of remaining dependence over that lag range, while the lag-30 p-value was 0.0543 and therefore slightly above the conventional 5% threshold.

The residual diagnostics are therefore mixed rather than uniformly satisfactory. The selected structure controlled the principal one-day and seven-day correlations effectively, but the lag-14 result indicates that some temporal pattern remained unexplained. The model provides a stronger adjusted-association specification than the OLS baseline, but it should not be presented as a fully specified or independently validated forecasting model.

## **B.5 Cross-Model Diagnostic Summary**

The three preferred models used the same daily and weekly autoregressive error structure but retained different fixed-effect calendar specifications. The total-demand and registered-user models used working-day status, whereas the casual-user model used separate weekday and holiday terms. Model selection was conducted independently for each response rather than by imposing one common specification on all three rental measures.

All three models converged successfully, and their estimated daily and weekly autoregressive parameters were statistically supported. The daily AR(1) estimates were 0.4611 for total demand, 0.3676 for casual demand, and 0.5168 for registered demand. The corresponding weekly seasonal AR(1) estimates were 0.1109, 0.2523, and 0.1012. These estimates describe dependence remaining in the model errors after adjustment for the fixed effects; they do not represent direct effects of observed rentals on previous days.

After the autoregressive components were included, the residual correlations at the targeted one-day and seven-day lags were close to zero in all three models. The lag-1 residual correlations were  $-0.0428$  for total demand,  $0.0124$  for casual demand, and  $-0.0259$  for registered demand. The corresponding lag-7 correlations were  $-0.0101$ ,  $-0.0416$ , and  $-0.0122$ . Durbin–Watson statistics of 2.0847, 1.9702, and 2.0513 were also consistent with limited remaining first-order residual autocorrelation.

The broader Ljung–Box results were less uniformly satisfactory. For total demand, the p-values at lags 7, 14, and 30 were 0.0117, 0.0011, and 0.0057, respectively, indicating remaining joint serial dependence across each examined lag range. For casual demand, all three p-values were below 0.001, providing the clearest evidence of residual temporal structure not captured by the selected specification.

The registered-user diagnostics were mixed. The Ljung–Box p-value was 0.1532 at lag 7, providing no evidence of remaining joint serial dependence across the first seven lags at the 5% significance level. The lag-14 p-value was 0.0169, indicating remaining dependence over that range, while the lag-30 p-value was 0.0543 and therefore slightly above the conventional 5% threshold.

These results show why individual residual correlations and Durbin–Watson statistics should not be evaluated in isolation. Although the residual correlations at lags 1 and 7 were small, the Ljung–Box tests assessed dependence jointly across multiple lags and identified temporal structure that remained elsewhere in the residual sequence.

AIC and BIC values must not be used to rank the three preferred models against one another because the models were fitted to different response variables and likelihood scales. The lower AIC or BIC of one response model does not imply that it is more accurate or better specified than a model for another response. Information-criterion comparisons were used only among candidate specifications fitted to the same response.

Overall, the preferred models provide a substantial improvement over the OLS baselines for explanatory analysis by accounting explicitly for important daily and weekly temporal dependence. However, none of the three models achieved uniformly clean residual diagnostics across all examined lag ranges. Their coefficients and uncertainty estimates should therefore be interpreted conditionally and with appropriate caution.

The modeling workflow used the full historical sample for estimation and diagnostic assessment and did not include rolling-origin or other out-of-sample forecast validation. The models are consequently appropriate for reporting adjusted historical associations and supporting system-level planning, but they should not be described as fully validated forecasting models.

## Appendix C — Reproducibility and Project Deliverables

The project was organized as a structured analytical package so that the source data, transformations, analysis code, statistical outputs, figures, and final report could be traced separately. Raw source files were preserved without modification, while cleaned data, analytical outputs, and reporting materials were stored in dedicated directories.

The reproducibility package documents the sequence of work from data audit through final communication. Python notebooks contain the principal data-processing, exploratory, modeling, and visualization workflows. JMP scripts and saved reports document the supplementary statistical analysis conducted in JMP. Structured CSV outputs preserve the numerical results used in the management report.

The project organization supports analytical traceability: reported findings can be linked to specific datasets, notebooks, scripts, tables, figures, or diagnostic outputs. Reproducibility remains dependent on access to compatible software, required Python packages, and the documented project files. The package should therefore be understood as a transparent and reproducible analytical workflow rather than a fully automated production pipeline.

### C.1 Project Directory Structure

The project directory separates source data, transformed data, analytical code, statistical outputs, reporting materials, documentation, and portfolio deliverables. This separation reduces the risk of modifying original records and makes the origin of each reported result easier to identify.

The raw-data directory contains the original daily and hourly source files and is treated as read-only. Cleaned datasets are stored separately so that parsing, labeling, and data-quality flags do not overwrite the source records. Analytical tables, figures, and diagnostic outputs are also separated from the code that generates them.

Python and JMP materials are maintained in dedicated directories. The numbered Python notebooks document the principal workflow in execution order, while the JMP directory preserves the supplementary scripts and saved statistical reports. The reporting and documentation directories contain the management report and the materials required to understand its analytical basis.

Table 9 summarizes the role of each principal project directory. For continued reproducibility, generated datasets, tables, figures, and diagnostics should be updated by rerunning the relevant analytical workflow rather than by manually changing the exported files.

*Table 9. Project directory structure and analytical role*

| Directory          | Primary contents                                 | Reproducibility role                                                   |
|--------------------|--------------------------------------------------|------------------------------------------------------------------------|
| 01_data/raw        | Original day.csv and hour.csv source files       | Preserves the unmodified analytical starting point                     |
| 01_data/clean      | Validated day_clean.csv and hour_clean.csv files | Stores prepared records without overwriting the source data            |
| 01_data/analytical | Derived analysis-ready datasets where required   | Separates analytical transformations from cleaned source-level records |



| Directory              | Primary contents                                                                                                      | Reproducibility role                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 02_python/notebooks    | Numbered Python notebooks covering audit, cleaning, exploratory analysis, time-series modeling, and reporting figures | Documents the principal analytical workflow in execution order       |
| 02_python/scripts      | Reusable Python scripts where notebook logic is separated into standalone functions or procedures                     | Supports reuse and reduces duplication of analytical logic           |
| 03_jmp                 | Numbered JMP scripts and saved statistical reports                                                                    | Preserves the supplementary JMP analysis and diagnostic evidence     |
| 04_outputs/tables      | Structured CSV summaries and model-result tables                                                                      | Retains the numerical evidence used in the report                    |
| 04_outputs/figures     | Final explanatory figures                                                                                             | Preserves report-ready visual outputs generated from validated data  |
| 04_outputs/diagnostics | Data-quality and model-diagnostic outputs                                                                             | Supports review of sensitivity checks and model adequacy             |
| 05_reports             | Management report in DOCX format and final PDF export                                                                 | Stores the principal communication deliverables                      |
| 06_documentation       | Supporting methodology, data-quality, and reproducibility documentation                                               | Explains analytical decisions, definitions, and limitations          |
| 07_portfolio           | Portfolio-facing project materials                                                                                    | Separates presentation assets from the underlying analytical package |

*Note: The directory structure distinguishes source files from generated analytical outputs. The presence of a directory identifies its intended project role; individual contents should be confirmed against the final submitted package.*

*Source: Author's project organization and reproducibility framework.*

## C.2 Core Analytical Workflow and Notebook Sequence

The core Python workflow consists of five sequential notebooks covering data audit, data preparation, exploratory analysis, time-series regression, and explanatory visualization. The numeric prefixes define the intended execution order because later notebooks depend on validated datasets or structured outputs produced during earlier stages.

The workflow begins with the raw daily and hourly source files. The audit notebook evaluates data structure, integrity, consistency, coverage, and documented quality concerns without modifying the source records. The cleaning notebook then creates validated working datasets while preserving the original observation counts and rental totals.

The exploratory-analysis notebook uses the cleaned daily and hourly datasets to produce descriptive summaries, user-segment comparisons, calendar and weather analyses, hourly-demand profiles, and sensitivity checks. Its structured CSV outputs provide the numerical basis for the descriptive findings and operational interpretations reported in the main document.

The time-series regression notebook uses the validated daily dataset. It reproduces the total-demand OLS baseline previously estimated in JMP, evaluates candidate autoregressive error

structures, selects separate preferred specifications for total, casual-user, and registered-user demand, and exports the final coefficient and diagnostic tables.

The final visualization notebook uses the validated clean datasets and the confirmed analytical results to generate the six numbered explanatory figures included in the management report. These figures are reporting outputs rather than independent analytical models; their values must remain consistent with the validated descriptive summaries.

The notebooks should be executed in numeric order when the project is reproduced in a fresh environment. Generated CSV and image files should not be modified manually. Any required change should be made in the relevant notebook and propagated by rerunning that notebook and any dependent downstream steps.

The final submitted package should contain one authoritative notebook for each sequence number. Draft, duplicate, or superseded notebook copies should be excluded from the final analytical package to prevent uncertainty about which version generated the reported results.

*Table 10. Core Python notebook sequence and generated outputs*

| Authoritative notebook | Analytical role                                                                                                                                                              | Principal generated outputs                                                                                                               |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 01_data_audit.ipynb    | Audits structure, integrity, categories, dates, counts, hourly completeness, cross-dataset consistency, environmental boundaries, and data readiness                         | data_quality_summary.csv;<br>data_readiness_summary.csv; issue_register.csv;<br>incomplete_hourly_dates.csv;<br>missing_hours_by_date.csv |
| 02_data_cleaning.ipynb | Creates working copies, parses dates and timestamps, adds documented quality flags and readable labels, sorts records, and validates preservation of observations and totals | day_clean.csv; hour_clean.csv                                                                                                             |

| Authoritative notebook              | Analytical role                                                                                                                                                               | Principal generated outputs                                                                                                                                                              |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03_exploratory_data_analysis.ipynb  | Produces daily and hourly descriptive analysis, segment and calendar comparisons, weather analysis, and sensitivity checks                                                    | Structured EDA summary CSVs; hourly-demand and user-mix summaries; peak_hour_stability.csv; hourly_demand_sensitivity_summary.csv; humidity_sensitivity_summary.csv; exploratory figures |
| 04_time_series_regression.ipynb     | Reproduces the total-demand OLS baseline, evaluates daily and weekly autoregressive structures, selects response-specific preferred models, and performs residual diagnostics | 01_preferred_model_summary.csv through 06_preferred_model_diagnostics.csv                                                                                                                |
| 05_explanatory_visualizations.ipynb | Generates the six final management-report figures and verifies the values supporting their principal messages                                                                 | 01_annual_growth_and_segment_contribution.png through 06_demand_by_weather_condition.png                                                                                                 |

*Note: Notebook names refer to the authoritative versions retained in the final submitted package. Generated outputs should be reproduced from the documented code rather than modified manually.*

*Source: Author's Python notebooks and validated project outputs.*

### C.3 Statistical Software, Environment, and Execution Requirements

The analytical workflow used Python as the principal environment for data audit, cleaning, exploratory analysis, time-series regression, diagnostic evaluation, and figure generation. JMP was used for supplementary distributional analysis, regression exploration, the original ordinary least squares baseline, and residual review.

The preferred autoregressive models were estimated and diagnosed in Python. JMP and Python were compared only for the total-demand OLS baseline using the same predictors and sum-coding convention. The project should therefore not be described as having reproduced the final autoregressive models independently in both software environments.

The Python notebooks require a Jupyter-compatible environment and the packages imported by the final notebook versions. The workflow relies principally on tabular data processing, numerical computation, statistical modeling, diagnostic testing, and static figure generation. The definitive dependency list and software versions should be obtained directly from the environment used for the final successful execution rather than reconstructed from memory.

All notebooks assume that the documented project-directory structure and relative file locations are preserved. The workflow should be launched from the project root, or from an environment in which the relative paths resolve to the same raw-data, clean-data, output, and report directories. Changing folder names or moving individual files without updating the code may break the workflow or cause outputs to be written to unintended locations.

The raw source files should remain unchanged. Generated datasets, analytical tables, diagnostics, and figures should be recreated by executing the relevant notebook rather than edited directly. Before reproduction, the required output directories should exist or be created by the workflow, and the user should confirm that Python has permission to read from the data directories and write to the output directories.

The notebooks should be run in numeric order. A successful reproduction should verify that the expected observation counts, rental totals, data-quality findings, principal descriptive statistics, selected model summaries, residual diagnostics, and six final figures agree with the validated outputs used in this report.

The reported workflow does not depend on a stochastic simulation, sampling procedure, or randomized train-test split. A random seed is therefore not a documented requirement for reproducing the reported results. Any future extension involving resampling, cross-validation, simulation, or machine-learning procedures should record and fix the relevant random seed.

For long-term reproducibility, the final submission should include an environment record generated from the working Python environment, together with the JMP version used for the supplementary analysis. Suitable environment records include a requirements file, a Conda environment export, or an equivalent package manifest. The recorded environment should reflect the final successful execution of the authoritative notebooks.

*Table 11. Software roles and reproduction requirements*

| Component           | Role in the project                                                                              | Reproduction requirement                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Python and Jupyter  | Principal environment for audit, cleaning, EDA, time-series regression, diagnostics, and figures | Use the final notebook versions and the packages imported by them                           |
| JMP                 | Supplementary exploration, OLS estimation, and residual review                                   | Record the JMP version and retain the final scripts and saved reports                       |
| Project directories | Control the relative locations of data, code, outputs, and reports                               | Preserve the documented folder structure or update all paths consistently                   |
| Raw data            | Provide the unchanged analytical starting point                                                  | Keep day.csv and hour.csv unmodified                                                        |
| Generated outputs   | Preserve validated datasets, tables, diagnostics, and figures                                    | Regenerate from code rather than edit manually                                              |
| Environment record  | Documents software and package dependencies                                                      | Export from the environment used for the final successful run                               |
| Execution order     | Maintains dependencies among workflow stages                                                     | Run notebooks 01 through 05 in numeric order                                                |
| Validation checks   | Confirm successful reproduction                                                                  | Compare key counts, totals, findings, model diagnostics, and figures with validated outputs |

*Note: Exact Python package and software-version numbers should be copied from the environment used for the final successful execution. They should not be estimated or reconstructed from memory.*

*Source: Author's documented analytical workflow and software use.*

## C.4 Validated Reporting Tables and Structured Outputs

The analytical workflow produced structured CSV outputs so that the numerical evidence used in the report could be reviewed independently of the notebooks and figures. These files preserve the validated descriptive summaries, preferred-model estimates, residual diagnostics, and documented data-quality checks.

Only outputs that support a reported finding, model interpretation, sensitivity result, or reproducibility requirement should be treated as authoritative deliverables. Intermediate files created during exploration may remain in the working environment, but they should not be presented as final reporting outputs unless their purpose is documented.

The descriptive reporting outputs contain the values used to support the principal management findings, including annual growth, monthly variation, hourly peak timing, user composition by day type, weekday casual demand, and weather-related differences. Additional hourly and day-type summaries support the operating-window recommendations.

The preferred-model outputs form a separate reporting set. They contain the selected model specifications, common adjusted relationships, calendar effects, temporal parameters, complete parameter estimates, and residual diagnostics for total, casual-user, and registered-user demand.

Data-quality and sensitivity outputs document the evidence behind the two issue-register entries and the checks used to evaluate their analytical influence. Missing date-hour combinations were not converted to zero-demand observations, and the zero-humidity records were retained rather than replaced. The corresponding structured files preserve the issue definitions, affected dates, and sensitivity results.

Table 12 identifies the authoritative structured outputs retained for the final reporting package. The inventory is intentionally limited to files with a clear analytical or reporting role; it is not intended to catalogue every intermediate object generated during exploration.

*Table 12. Authoritative structured analytical outputs*

| Output group                           | Authoritative files                                                                                                                                                                                              | Principal reporting role                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Descriptive management summaries       | annual_growth_summary.csv; monthly_demand_summary.csv; peak_hour_summary.csv; segment_mix_by_day_type.csv; weekday_demand_summary.csv; weather_condition_summary.csv                                             | Supports the six principal descriptive findings and management figures                      |
| Hourly and operating-profile summaries | peak_hour_by_day_type.csv; hourly_demand_by_day_type.csv; hourly_segment_mix_by_day_type.csv; operational_implications.csv                                                                                       | Supports day-type, segment-specific timing, and operating-window interpretations            |
| Preferred-model reporting outputs      | 01_preferred_model_summary.csv;<br>02_common_effect_estimates.csv; 03_calendar_effects.csv;<br>04_time_structure_parameters.csv;<br>05_all_preferred_model_parameters.csv;<br>06_preferred_model_diagnostics.csv | Supports model selection, adjusted relationships, parameter tables, and residual assessment |

| Output group                         | Authoritative files                                                                                                                                                                                                                | Principal reporting role                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Data-quality and sensitivity outputs | data_quality_summary.csv; data_readiness_summary.csv; issue_register.csv; incomplete_hourly_dates.csv; missing_hours_by_date.csv; humidity_sensitivity_summary.csv; hourly_demand_sensitivity_summary.csv; peak_hour_stability.csv | Documents audit findings, affected records, analytical treatment, and sensitivity evidence |

*Note: The inventory includes the principal structured files used to support the final report. Exploratory or duplicate outputs should be excluded from the submitted reporting package unless they have a separately documented analytical purpose.*

*Source: Author's validated Python outputs and final reporting workflow.*

## C.5 Final Presentation Figures

The final management report contains six explanatory figures generated by the authoritative visualization notebook. Each figure was designed to communicate one principal management finding and was created from validated daily or hourly analytical data.

The figures are reporting outputs rather than independent sources of evidence. Their values were checked against the corresponding structured CSV summaries before inclusion in the report. Titles, labels, units, legends, and category names were selected to remain consistent with the terminology used throughout the analysis.

The final image files should be regenerated from 05\_explanatory\_visualizations.ipynb whenever the underlying data, calculations, or visual specifications change. They should not be edited manually in a way that alters plotted values, category order, labels, or analytical meaning.

Table 13 documents the six authoritative figure files, their principal messages, and the structured outputs used to validate them. The figure numbers correspond to their order in the main management report.

*Table 13. Final presentation figures and validation basis*

| Figure and authoritative file                            | Principal validated message                                                                                        |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Figure 1 — 01_annual_growth_and_segment_contribution.png | Average daily rentals were 64.4% higher in 2012, and registered users accounted for 84.5% of the observed increase |
| Figure 2 — 02_monthly_demand_trend.png                   | Average demand in 2012 exceeded the corresponding 2011 level in all 12 calendar months                             |
| Figure 3 — 03_hourly_segment_profiles.png                | Casual demand peaked at 14:00, while registered-user and total demand peaked at 17:00                              |
| Figure 4 — 04_user_mix_by_day_type.png                   | Registered users represented 86.8% of working-day rentals and 68.3% of non-working-day rentals                     |
| Figure 5 — 05_casual_demand_by_weekday.png               | Average weekend casual demand was 124.3% higher than the Monday-to-Friday average                                  |

| Figure and authoritative file                 | Principal validated message                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Figure 6 — 06_demand_by_weather_condition.png | Average completed rentals declined across less favorable observed weather categories; light rain or snow was represented by only 21 daily observations |

*Note: The figures summarize validated descriptive evidence and should not be interpreted as causal estimates. Figure 6 does not support conclusions about the most severe weather category because no daily observations were available for that category.*

*Source: Author's final visualization notebook and validated descriptive-output files.*

## C.6 Final Submission Package and Quality-Control Checks

The final submission package should contain only the authoritative files required to review, reproduce, and communicate the completed analysis. Draft, duplicate, temporary, or superseded files should be removed from the submitted package or stored outside the final project directory.

The analytical package should retain the original daily and hourly source files, the validated clean datasets, the five numbered Python notebooks, the final JMP scripts and saved reports, the authoritative structured CSV outputs, the six final explanatory figures, and the completed management report in both DOCX and PDF formats.

The DOCX file is the editable master report. The PDF is the controlled presentation version and should be generated only after the Word document has passed its final content and formatting review. Both versions should contain the same text, tables, figures, captions, numbering, and appendix content.

Before submission, the five Python notebooks should be reviewed in numeric order to confirm that they refer to the documented project paths and generate the expected outputs. The final package should contain only one authoritative version of each numbered notebook. Generated CSV and image files should agree with the values and messages reported in the management document.

The final Word quality-control review should verify the table of contents, list of figures, list of tables, heading hierarchy, automatic captions, cross-references, page orientation, section breaks, page numbering, table continuation settings, image quality, and consistency of terminology. All automatic fields should be updated before the final PDF is created.

The PDF should then be inspected page by page. Particular attention should be given to tables that continue across pages, repeated table headers, landscape sections, figure resolution, caption placement, isolated headings, unexpected blank pages, and any text or table content that extends beyond the page margins.

The final package should also be checked for analytical consistency. Observation counts, rental totals, reported percentages, peak hours, model estimates, diagnostic values, file names, and stated limitations should agree across the notebooks, structured outputs, figures, tables, and narrative report.

Completion of these checks supports a transparent and professionally controlled analytical submission. It does not imply that the workflow is a production system or that every future execution

will be identical under different software versions. The submitted package represents the authoritative historical analysis documented in this report.